PET VET

- FIND, BOOK, CARE... ALL IN ONE PLACE -

Final Report

Group No : 16

IS2201 Group Project

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1. Introduction

1.1 Domain Description

The Pet Care and Veterinary Management System (PETVET) is a comprehensive web-based platform designed to connect pet owners, clinics, vets, sitters, trainers, and shop sellers through a centralized system. It provides tools for appointment scheduling, medical records, pharmacy operations, shop sales, breeding and pet adoption listings, Lost & Found reporting, and location-based discovery. The system supports users with multiple roles (e.g., a user can act as both a trainer and a seller) through dynamic, module-based access.

Each registered veterinary clinic will have its own digital workspace within the system to manage bookings, staff, inventory, medical records, and sales, allowing them to operate independently as a clinic management platform.

1.2 Current System and Limitations

Most veterinary practices, pet shops and pet service providers still use manual or semi-automated systems for managing their day-to-day operations. Veterinary businesses often process appointments through phone or log-books, which complicates correct appointment records and scheduling management.

Key limitations of the current system include:

- Veterinary medical records are stored in physical files inaccessible by clients or remote vets
- Pet owners struggle to access health, vaccination and treatment records after visits
- Service providers (trainers, sitters, groomers) lack a unified platform to reach clients
- Manual billing systems are time-consuming, error-prone and expensive
- No communication mechanism between clients and clinics for timely confirmation
- No data protection or backup options, risking sensitive information
- No analytical reporting support for management insights

1.3 Goal & Objectives

Primary Goal: To develop a fully functional, secure, and user-friendly web application that supports pet health management, service booking, product purchasing, and clinic operations, accessible by multiple user roles with appropriate permissions.

Objectives:

- Develop a secure multi-role login system for administrators, clinic managers, veterinarians, receptionists, clients, sellers, sitters, groomers, and trainers
- Create an easy-to-use appointment booking system via online or over the phone
- Allow vets to manage pet medical records and issue prescriptions securely
- Enable clients to manage multiple pet profiles and view their health history and invoices
- Help independent vet clinics to manage all their services and functionalities
- Build a pharmacy module to manage medicine stock, prescriptions, and expiry tracking
- Add a pet shop feature where clients can browse and order products online
- Include a Lost & Found module to help owners post and find missing pets

1.4 Process Re-engineered

The system re-engineers the traditional manual appointment booking, medical records management, and shop inventory processes into a centralized, role-based digital platform. Physical appointment books are replaced with a time-slot-based online booking calendar. Paper-based medical records are

digitized and linked to pet profiles. Manual financial tracking is replaced by an automated billing and financial management module with commission-based income tracking.

1.5 Assumptions

- All developers have access to required technologies (HTML, CSS, JavaScript, PHP, MySQL) and development tools
- Users (vets, clients, receptionists) have basic digital literacy and can use web applications without in-depth training
- Each pet has a unique profile and is correctly linked to its respective client
- Each clinic, trainer, and sitter registered keeps their availability honestly in the system
- Project team members will be available throughout the development lifecycle

2. Feasibility Study

2.1 Technical Feasibility

Technical feasibility confirms that the required technologies and resources are suitable and sufficient to complete the project successfully. The system uses well-supported, basic web technologies: HTML, CSS, and JavaScript for user interfaces; PHP for server-side logic; and MySQL as the main database. A basic payment gateway and Google Maps API will be integrated for online payments and location-based services respectively.

2.2 Schedule Feasibility

Commitment Type	Hours
Weekday (Mon–Fri)	1 hour/day
Weekend (Sat–Sun)	4 hours/day
Weekly total per member	13 hours
Estimated duration	8 months (~32 weeks)
Total per member	416 hours
Total for team (4 members)	1664 hours

The project is designed to be completed within a 10-month period, with well-defined phases for requirements gathering, system design, development, testing, and final deployment.

2.3 Resource Feasibility

The team members possess technical skills in HTML, CSS, PHP, JavaScript, and MySQL. External services such as payment gateways offer free tiers suitable for academic projects. The project is guided by an experienced supervisor and co-supervisor.

2.4 Legal and Ethical Feasibility

The project complies with applicable laws and regulations including data protection and privacy laws. User data is handled with confidentiality. Role-based restricted access ensures data security. Ethical principles of transparency, accountability, and user privacy form the foundation of system design.

2.5 Social Feasibility

A Google Form survey was distributed to 52 potential users to gather feedback on preferred features, booking habits, and anticipated services. Results confirmed strong demand for the proposed system features including online appointment booking, medical records access, and pet care services.

3. Requirements

3.1 Stakeholders / Actors

The system supports the following 12 actors:

- Administrator – Full system management: users, services, content, global analytics
- Clinic Manager – Oversees clinic workspace: appointments, staff, pharmacy/shop inventory
- Veterinarian – Manages appointments, medical records, and prescriptions
- Receptionist – Handles walk-ins, bookings, and clinic queue
- Client (Pet Owner) – Manages pet profiles, books services, shops products
- Trainer – Lists training sessions, manages appointments
- Sitter/Boarder – Manages calendar and accepts bookings
- Groomer – Books grooming time slots
- Seller – Sells pets and manages listings
- Breeder – Posts breed listings for mating/adoption
- Unregistered User – Basic browsing and registration access

3.2 Functional Requirements

General System Requirements

- Secure multi-role login (Admin, Vet, Trainer, Client, etc.)
- Module-based role system with dynamic dashboards based on selected roles
- Support for multi-role users (a user can be both trainer and sitter simultaneously)
- Audit logging for significant actions (edits, cancellations)

Booking & Appointment Management

- Time-slot-based online, phone, and walk-in booking
- Queue management with check-in/check-out and arrival tracking
- Rescheduling and cancellations with reason logs
- Waitlist function for overbooked days

Medical Records

- Vets create and maintain records linked to appointments

- Diagnosis, symptoms, medication, treatment, and file uploads
- Pharmacy stock tracking, expiry management, and auto-invoicing from prescriptions

Pet Shop & Financial

- Admin manages product listings; clients browse, purchase, and track orders
- Automated billing, payment processing, refunds, and invoice download
- Commission tracking on shop sales and service bookings

3.3 Non-Functional Requirements

- Performance: System response within 3 seconds; homepage loads in 2 seconds
- Usability: Mobile-responsive UI with consistent navigation across all roles
- Security: Role-based access control, bcrypt password hashing, HTTPS, SQL injection prevention
- Reliability: 99% uptime; daily automated backups; recovery within 30 minutes
- Maintainability: Modular codebase, consistent style guides, comprehensive action logging

3.4 In-Scope

- Module-based dashboards for all 12 roles with permission-based feature access
- Isolated clinic panels for appointments, pharmacy, medical history, and staff management
- Appointment booking system with online payments
- Medical records, prescriptions, and vaccination management
- Centralized pet shop with order management and invoice download
- Breeding, Sitter, Grooming, Trainer, and Boarding module support
- Lost & Found section with image uploads and contact integration
- Pet adopting, selling, and educational resource pages

3.5 Out-of-Scope

- Mobile Vet Clinic and Home Visit Management with map-based coordination
- Real-Time Chat or Video Consultations (telemedicine)
- Automated SMS/Email notifications and appointment reminders
- Community Space / Pet Social Hub for sharing stories and photos

3.6 Constraints and Limitations

- Technology: System must be implemented without third-party frameworks (academic restriction)
- Time: All functional modules, testing, and documentation within the academic timeline
- User Roles: Each role is restricted to defined functionalities; only Admins access system-wide reports
- Data: Manual entry required; no real-time sync with external clinics; uploaded files limited to JPG/PDF under 5MB

4. System's Architecture

4.1 Components and Their Functionalities

The system follows an MVC (Model-View-Controller) architecture with four main component categories:

User Interfaces (View Layer)

Role-based UIs for Login, Admin, Clinic Manager, Vet, Receptionist, Client, Trainer/Sitter/Groomer, Seller/Breeder, and Unregistered User. Each UI communicates directly with its corresponding controller.

Controllers (Logic Layer)

Auth, Admin, Clinic, Vet, Receptionist, Client, Trainer, Sitter, Seller, and Finance controllers. They process user inputs, execute business logic, and coordinate data flow between Views and Models.

Models (Data Layer)

Admin, Breeder, Clinic Manager, Guest, Pet Owner, Sitter, Trainer, and Vet models. Each model interacts with the database to store, retrieve, and manage data while enforcing business rules.

Database (Central Data Repository)

MySQL database storing users, pets, appointments, medical records, services, listings, and financial transactions. Supports encrypted storage for sensitive information including credentials and payment data.

4.2 Component Interactions

The components interact through the MVC framework:

- UI → Controller: User actions are sent to the corresponding controller for validation and logic execution
- Controller → Model: The controller communicates with the model for CRUD operations (e.g., VetController uses VetModel)
- Model → Database: Models handle all database interactions using SQL queries
- External APIs: Payment Gateway, Google Maps API, and PHPMailer integrate with the backend layer

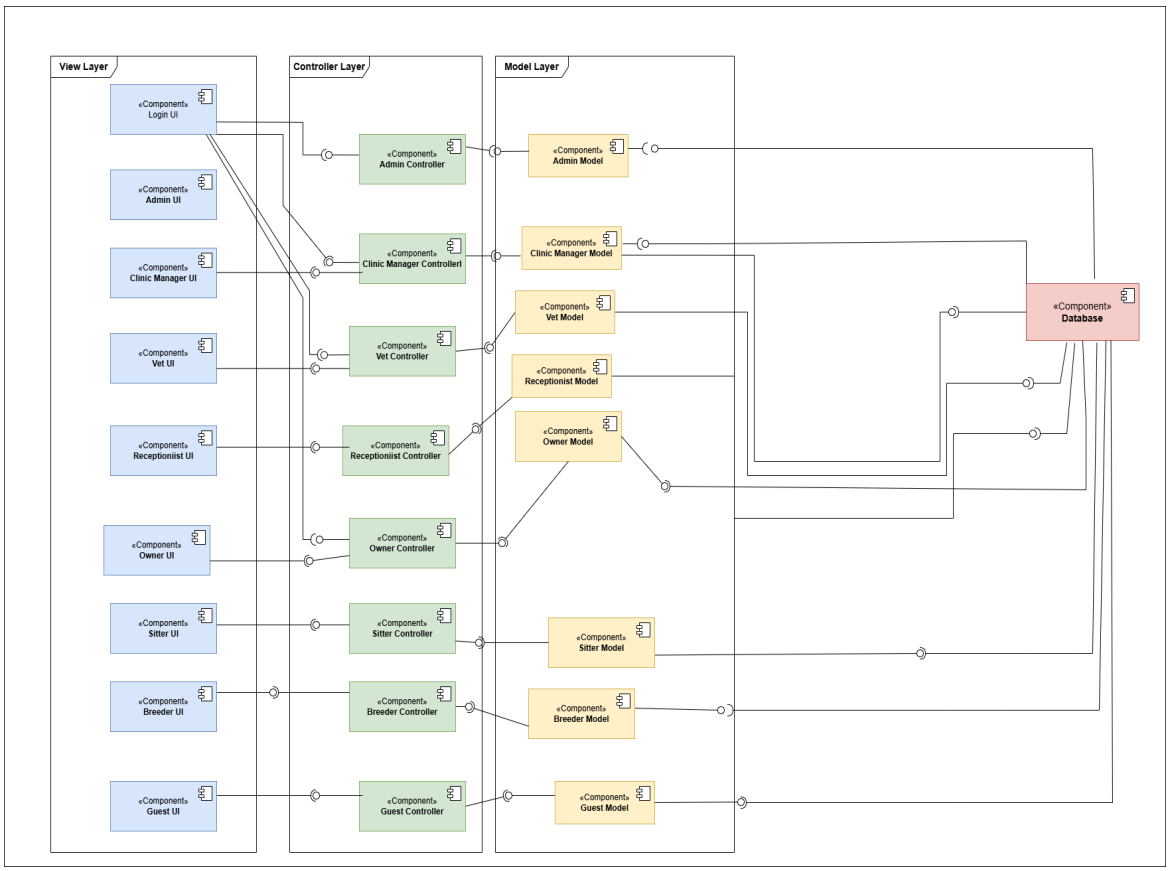

Figure 4.1 – System Component Interaction Diagram

5. System Design Diagrams

5.1 Use Case Diagrams

<u>Sitter</u>

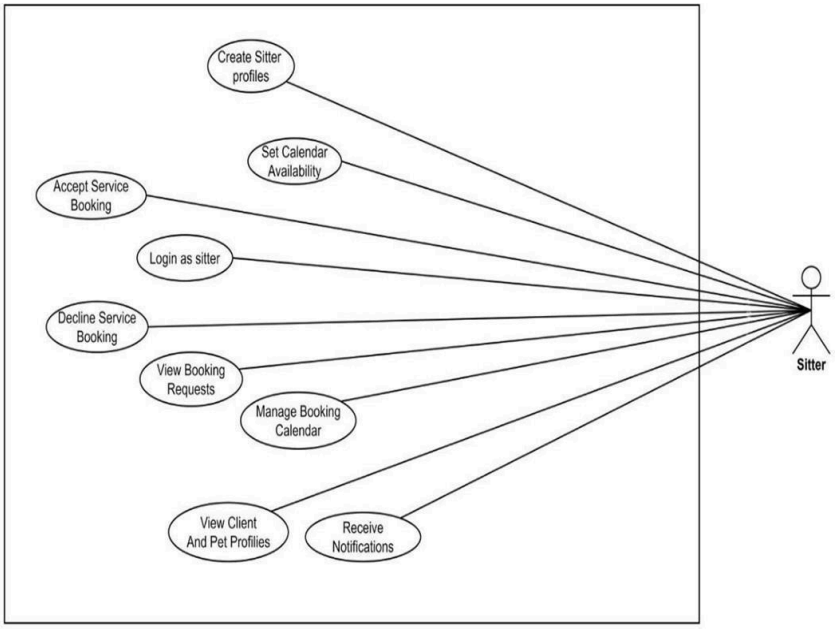

Figure 5.1.1 – Use Case Diagram: Sitter

Groomer

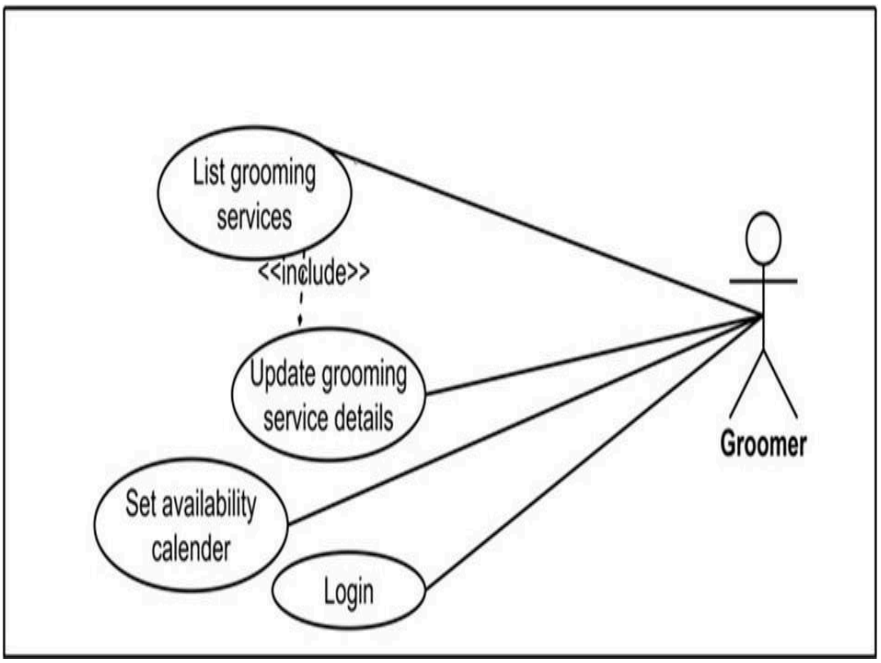

Figure 5.1.2 – Use Case Diagram: Groomer

Trainer

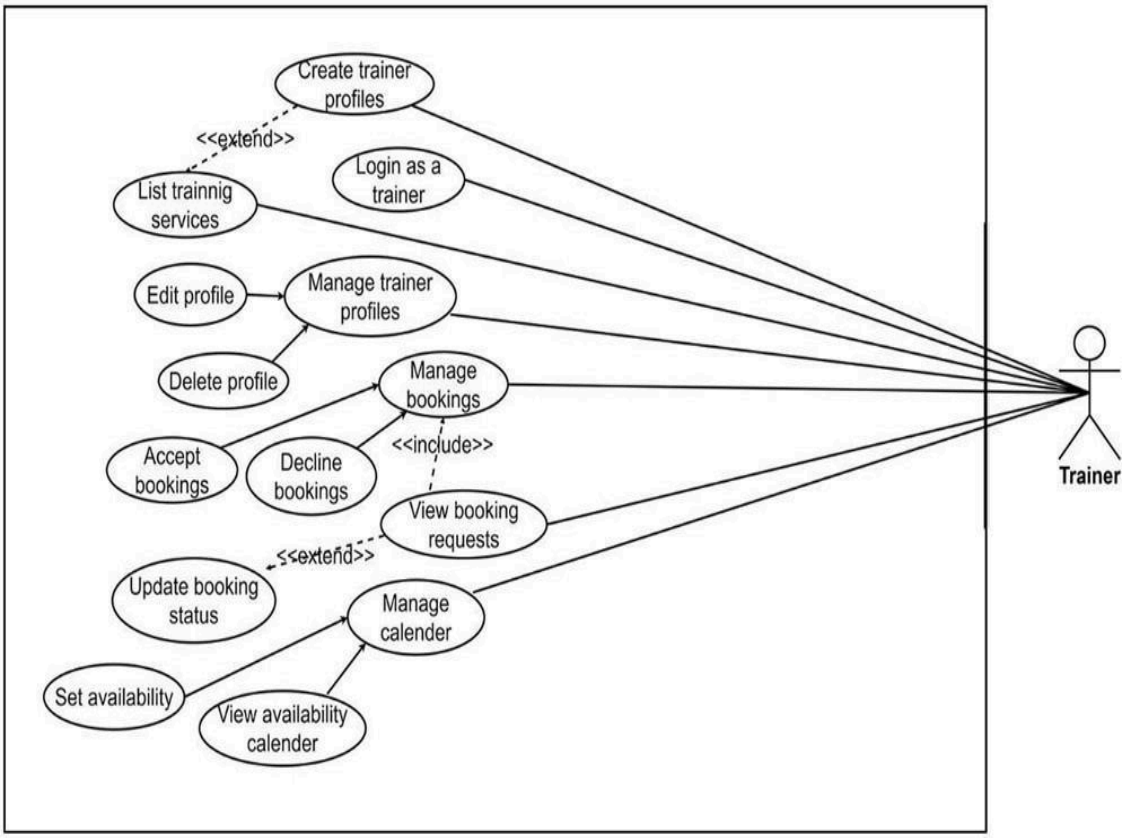

Figure 5.1.3 – Use Case Diagram: Trainer

Receptionist

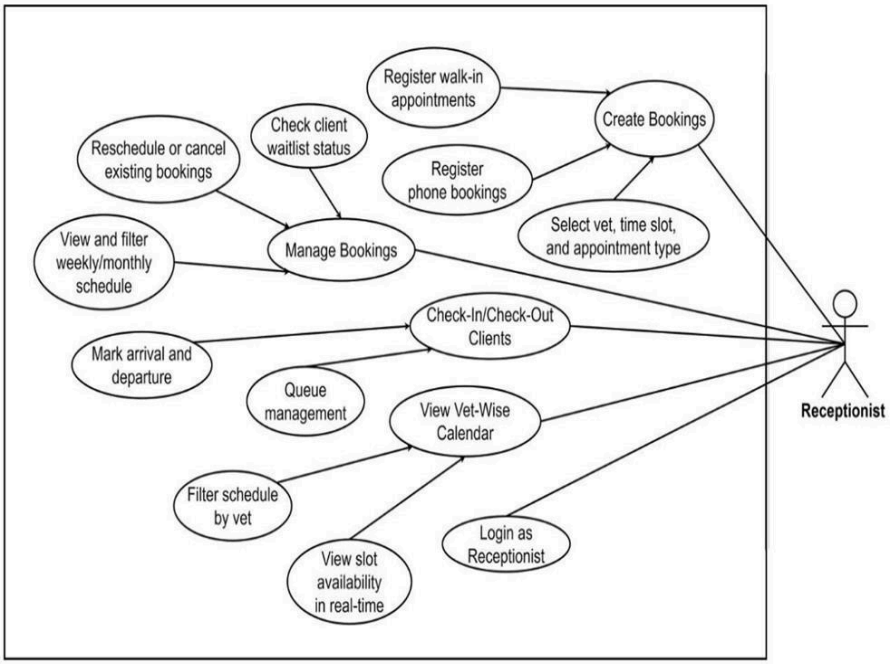

Figure 5.1.4 – Use Case Diagram: Receptionist

Vet

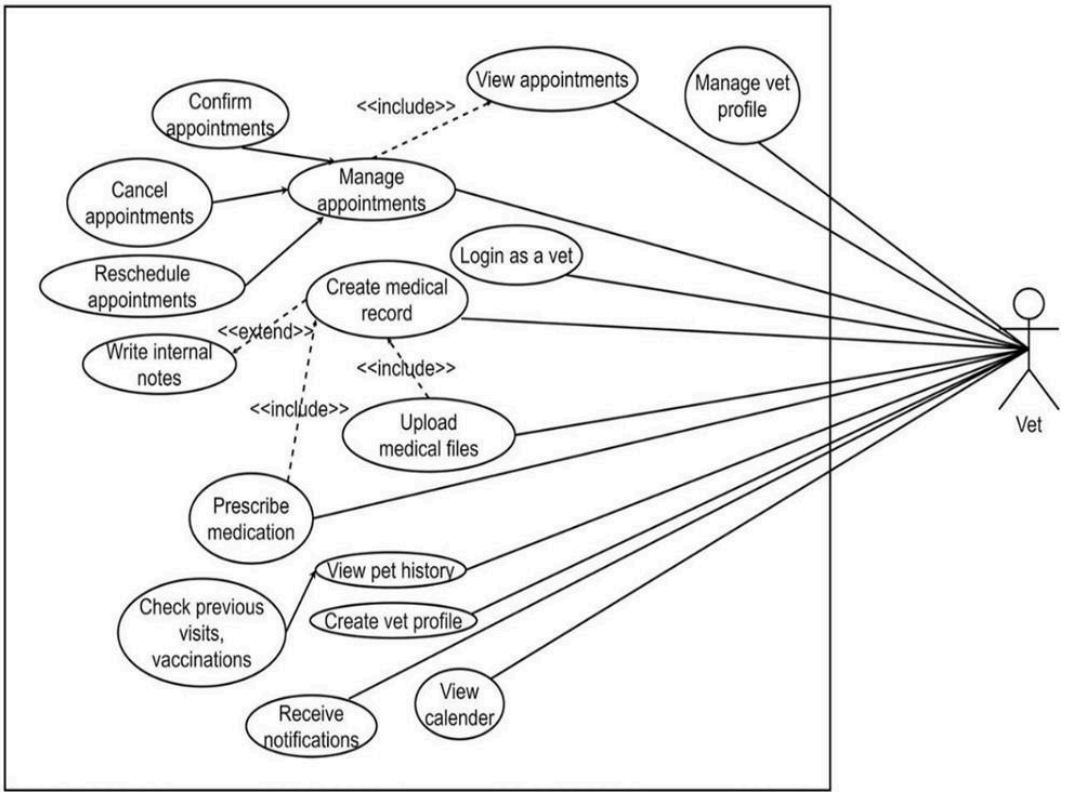

Figure 5.1.5 – Use Case Diagram: Vet

Clinic Manager

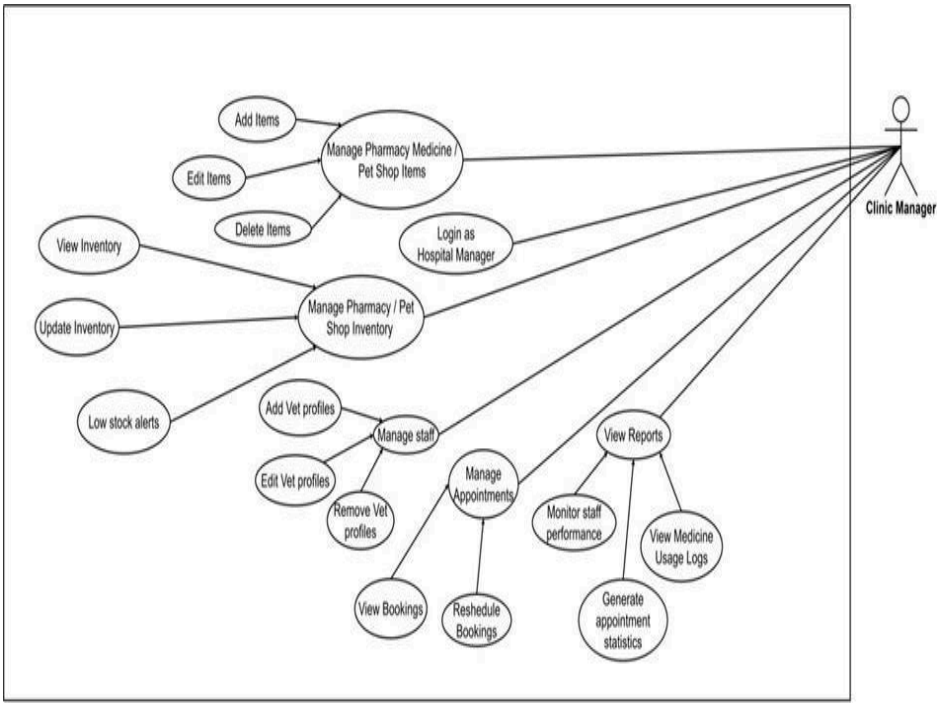

Figure 5.1.6 – Use Case Diagram: Clinic Manager

Admin

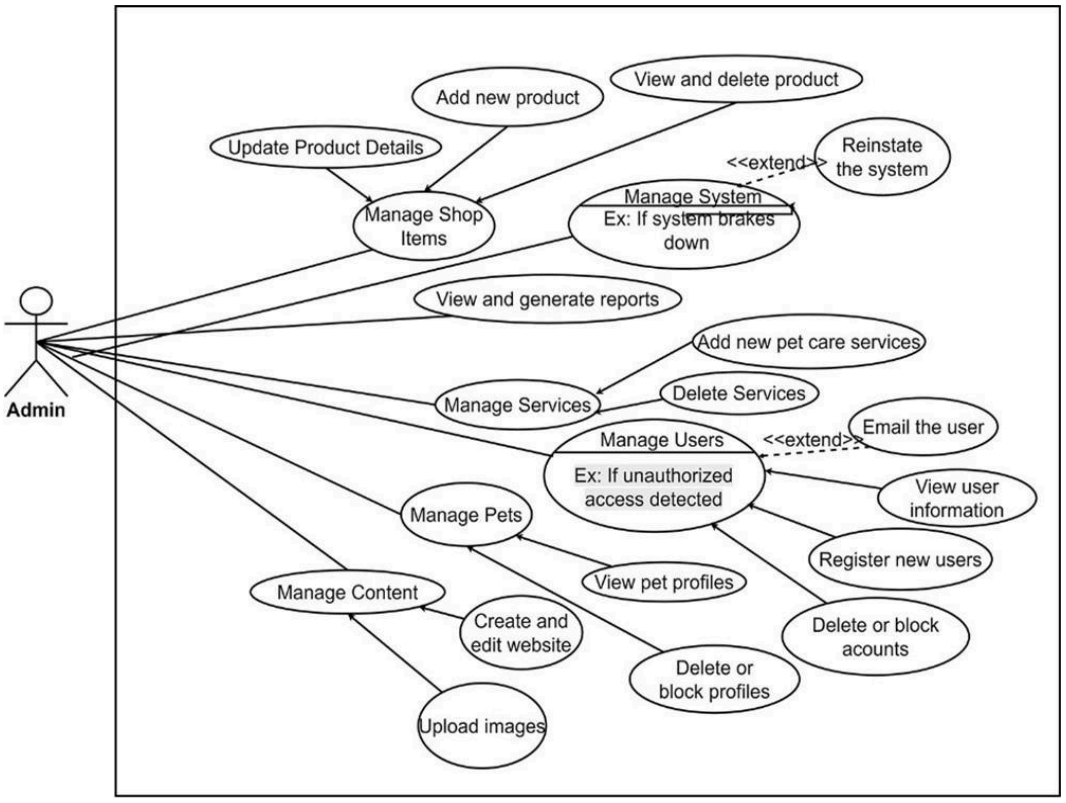

Figure 5.1.7 – Use Case Diagram: Admin

Client (Owner)

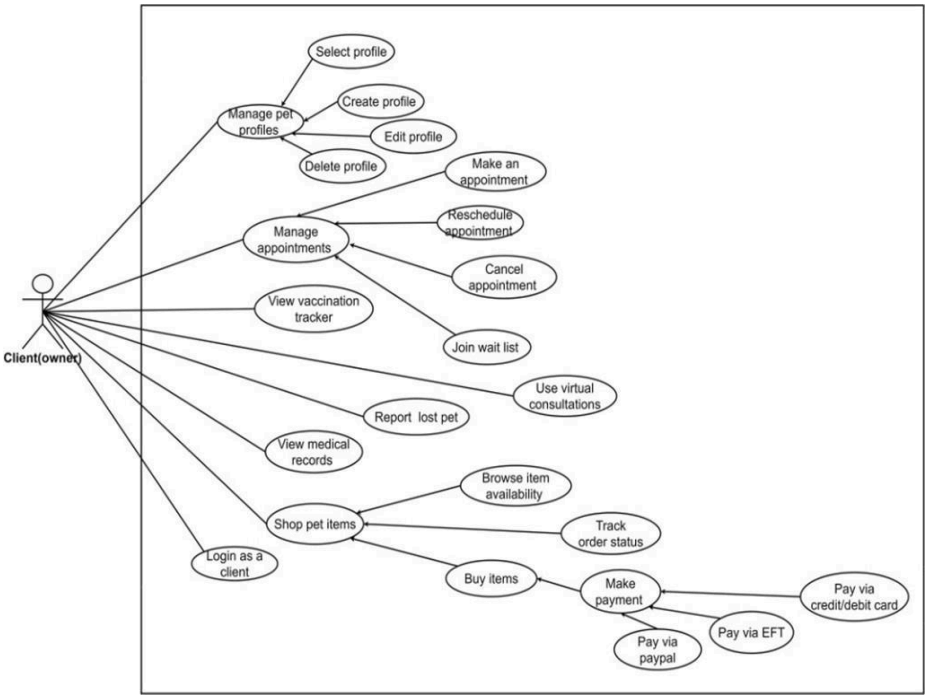

Figure 5.1.8 – Use Case Diagram: Client (Owner)

Unregistered User

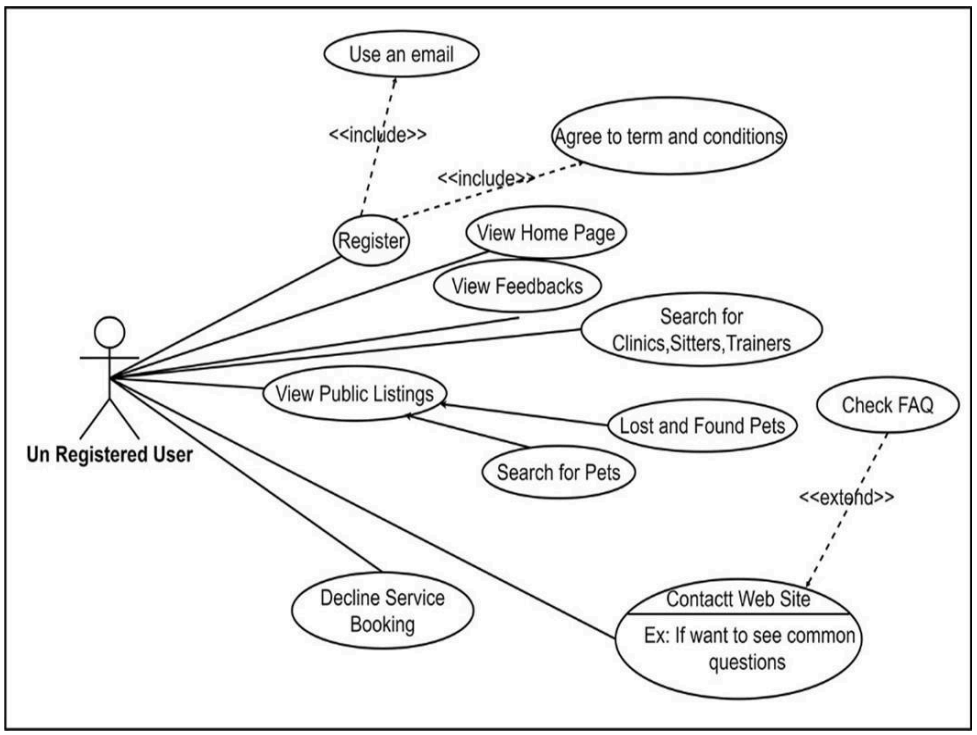

Figure 5.1.9 – Use Case Diagram: Unregistered User

Breeder

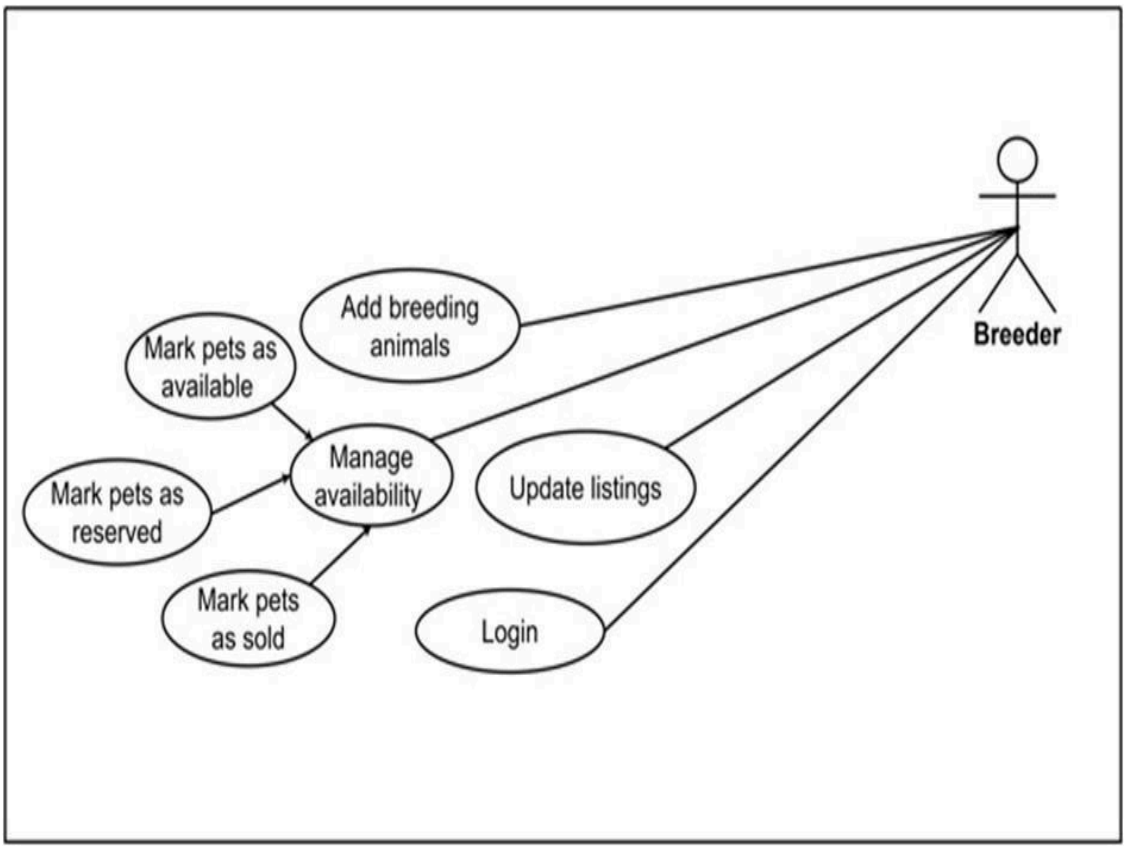

Figure 5.1.10 – Use Case Diagram: Breeder

5.2 Class Diagram

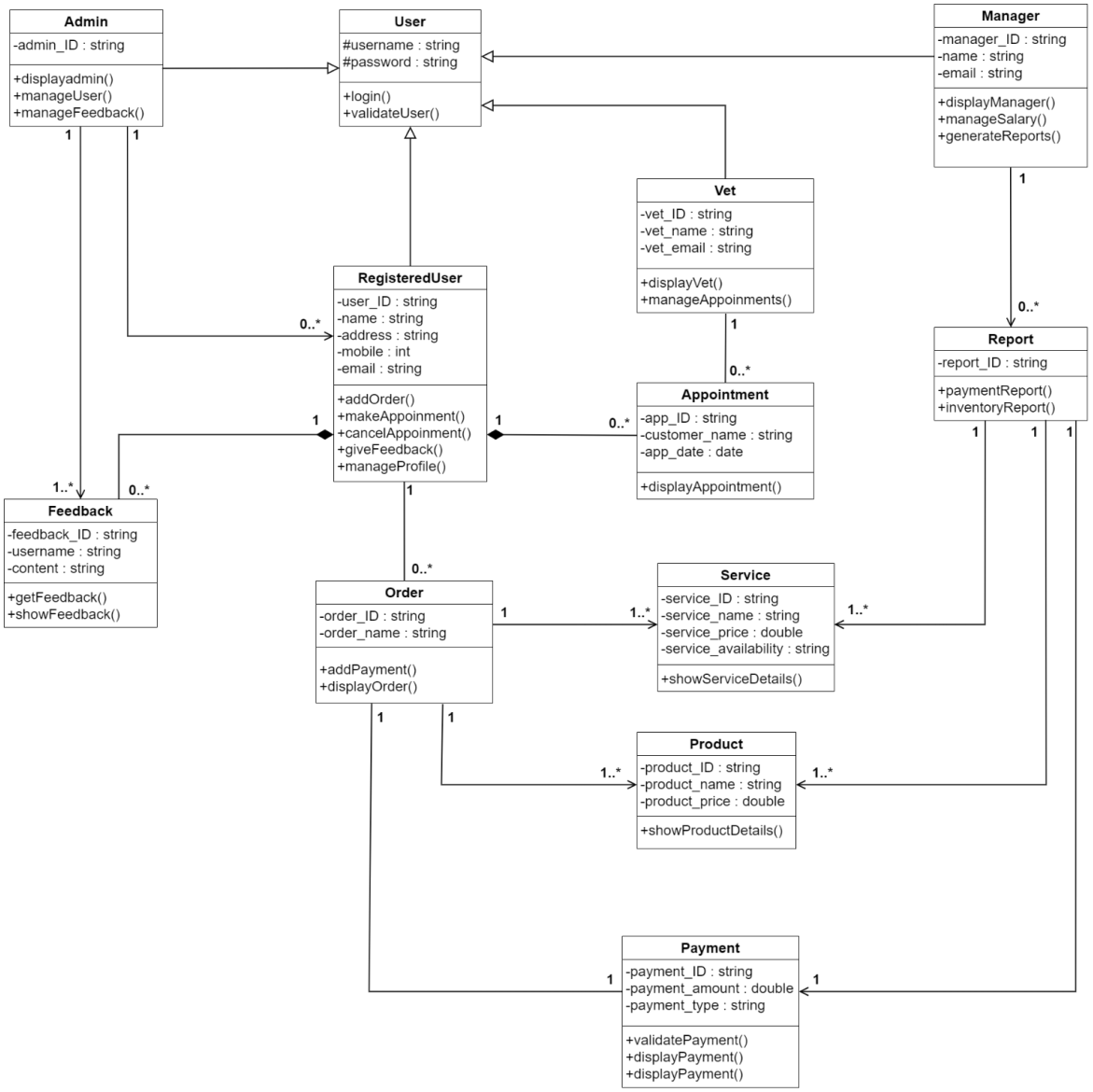

Figure 5.2 – Class Diagram

5.3 ER Diagram

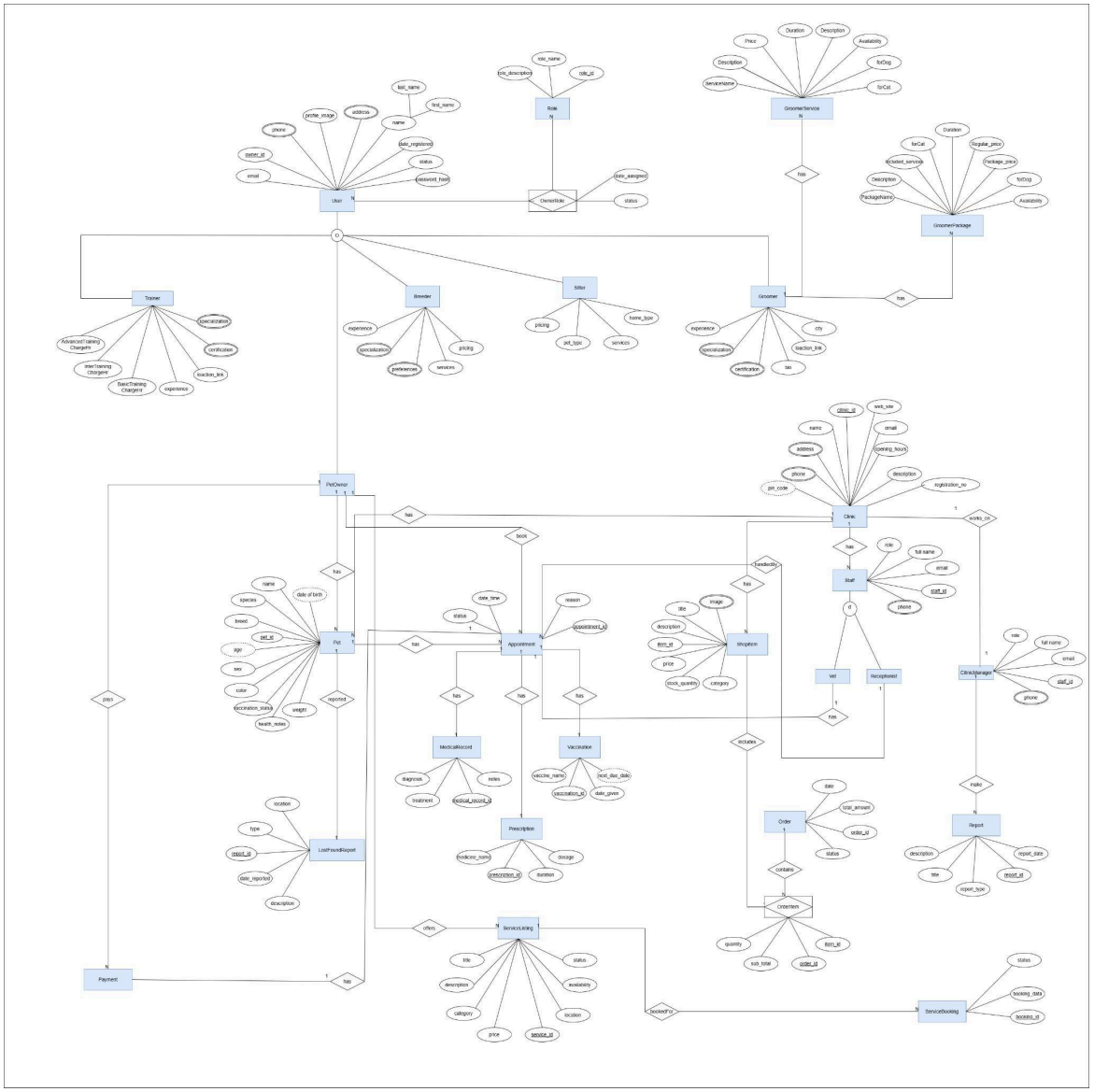

Figure 5.3 – Entity Relationship (ER) Diagram

5.4 Activity Diagrams

Vet Login and Dashboard Access

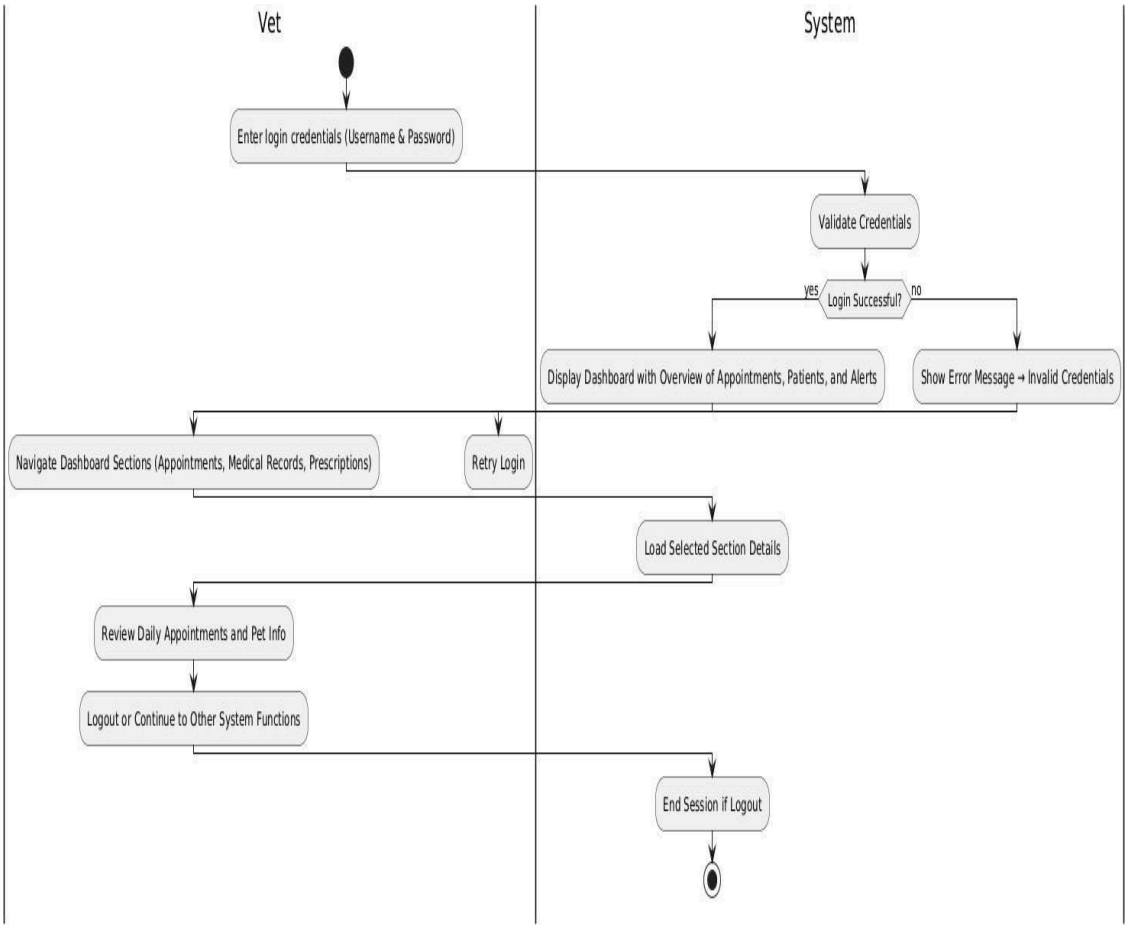

Figure 5.4.1 – Vet Login and Dashboard Access

Trainer Appointment Management

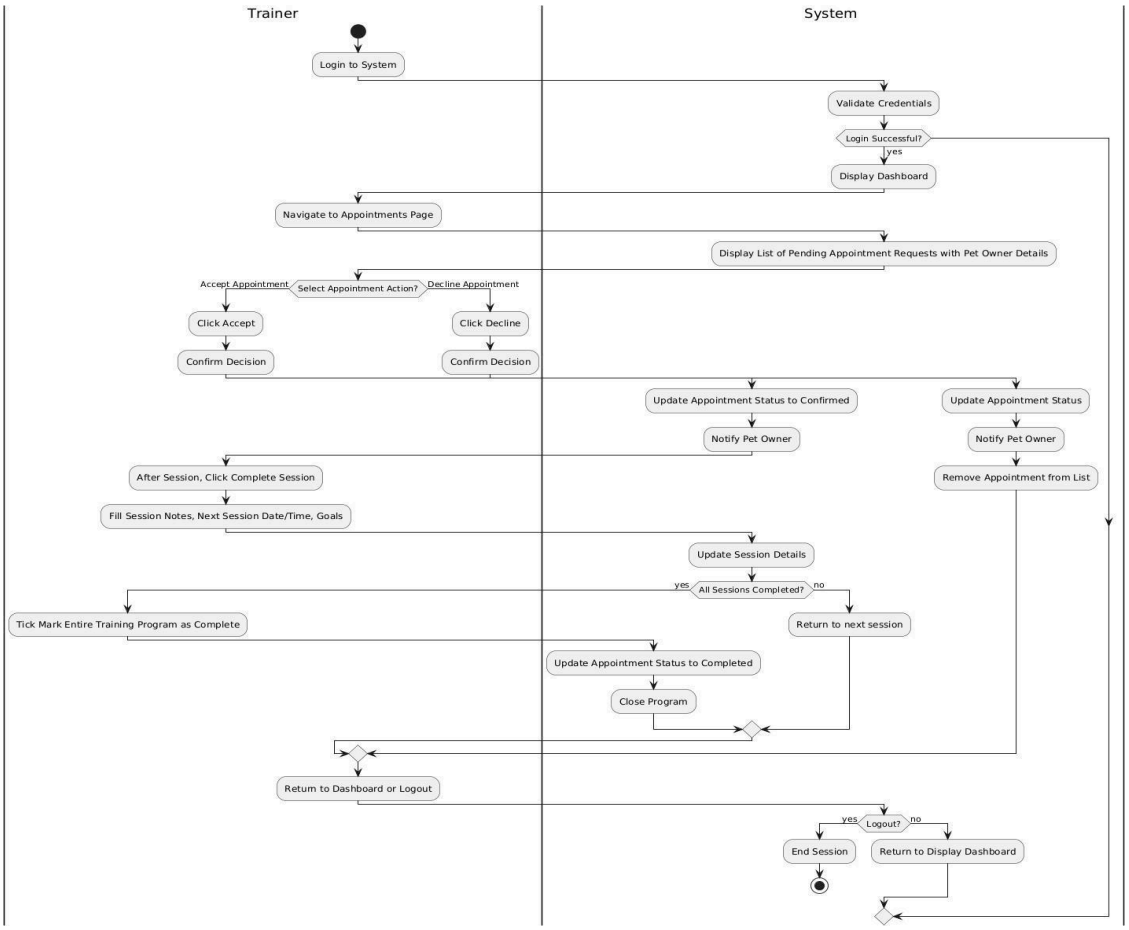

Figure 5.4.2 – Trainer Appointment Management

Vet Views Scheduled Appointments

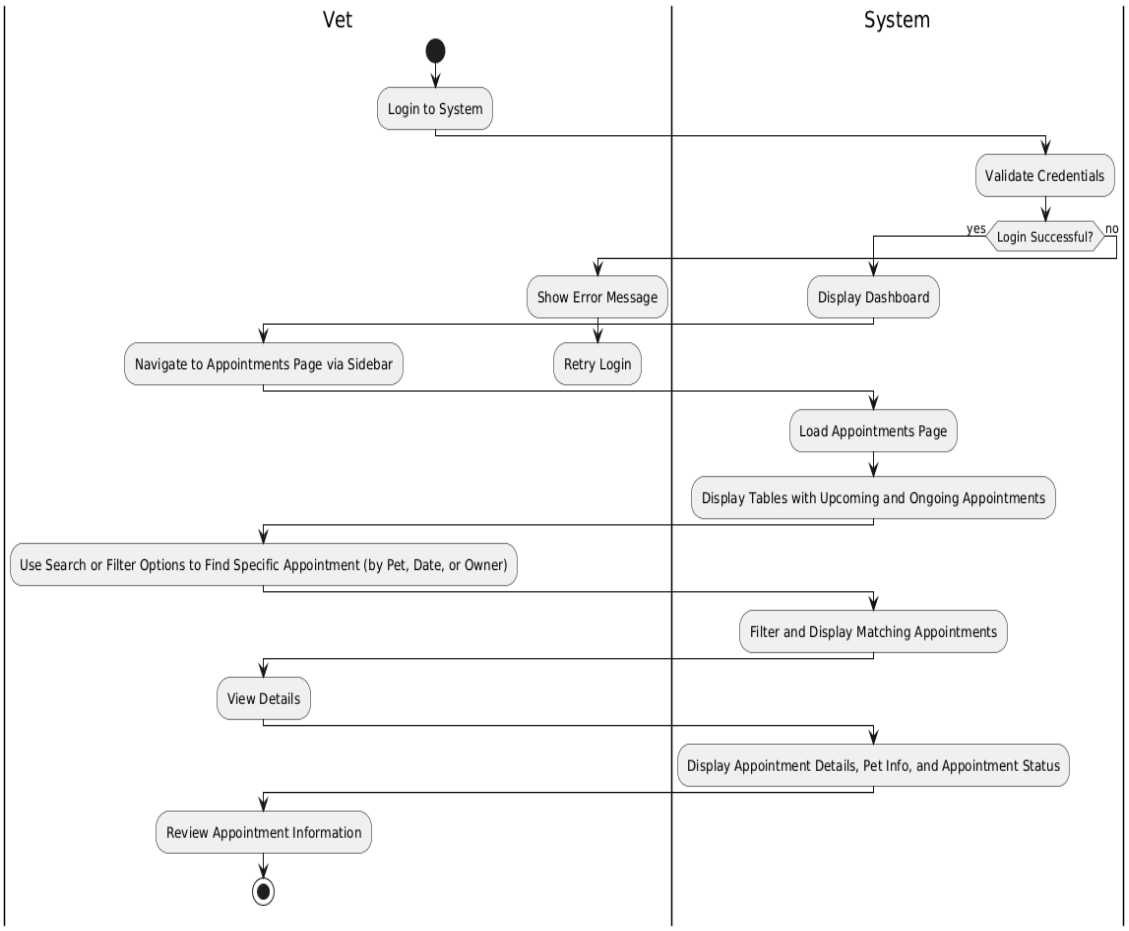

Figure 5.4.3 – Vet Views Scheduled Appointments

Vet Views Medical Records

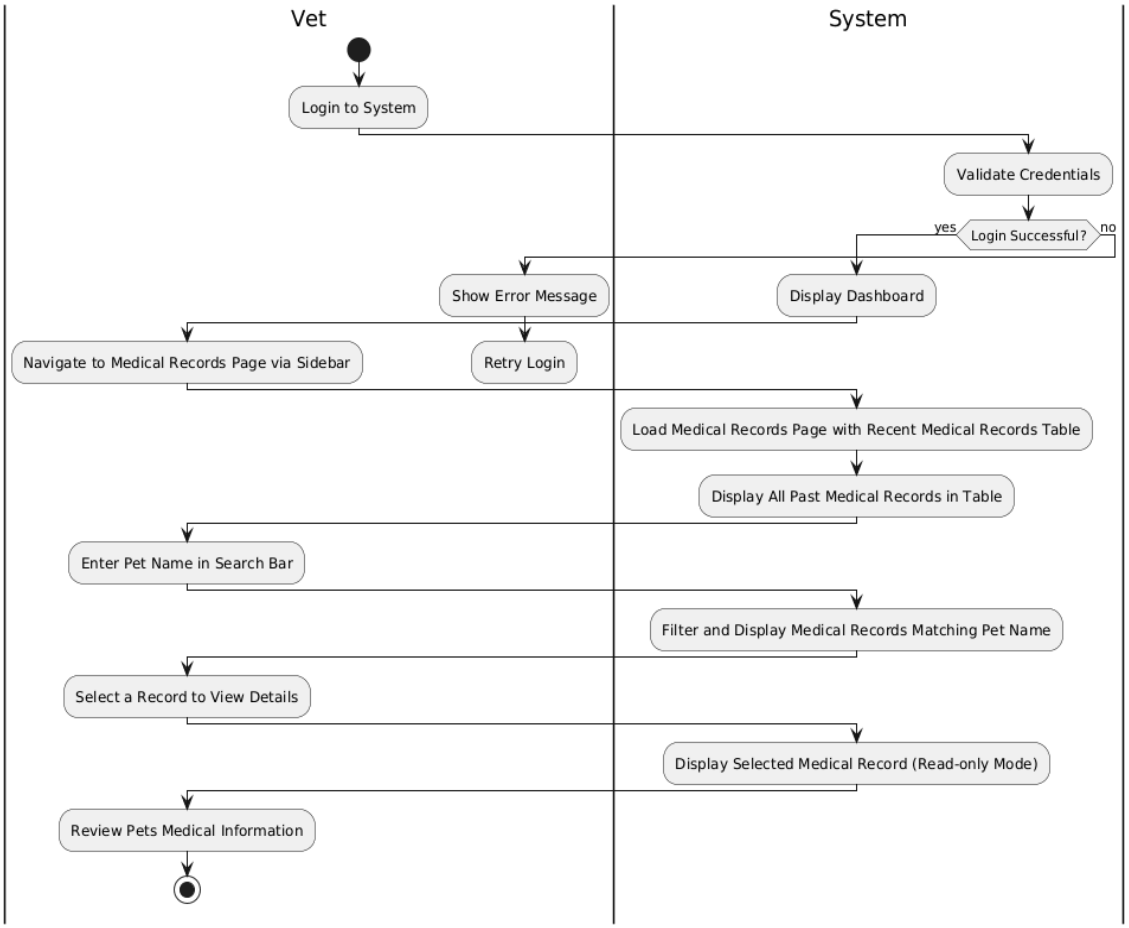

Figure 5.4.4 – Vet Views Medical Records

Vet Views Prescription Records

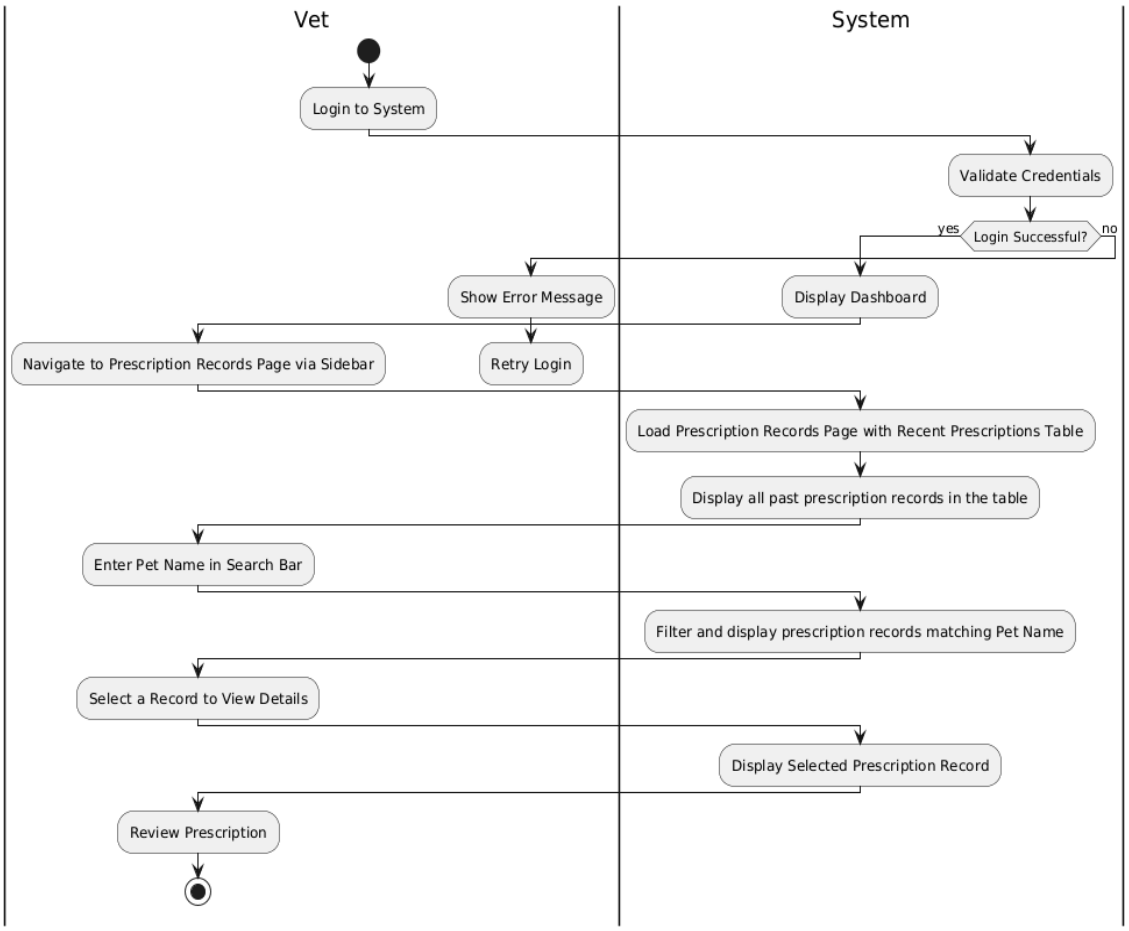

Figure 5.4.5 – Vet Views Prescription Records

Vet Views Vaccination Record

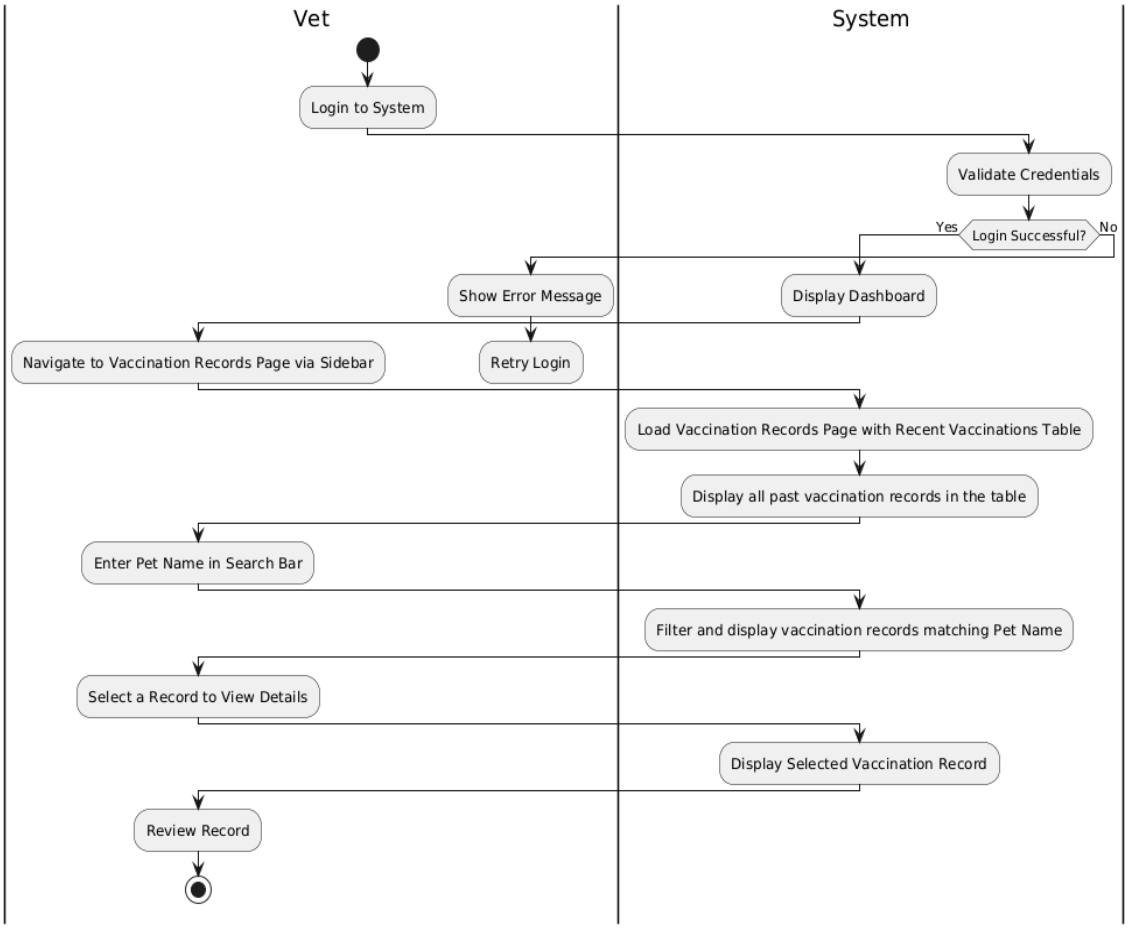

Figure 5.4.6 – Vet Views Vaccination Record

Vet Adds Medical Record

Figure 5.4.7 – Vet Adds Medical Record

Vet Adds Prescription Record

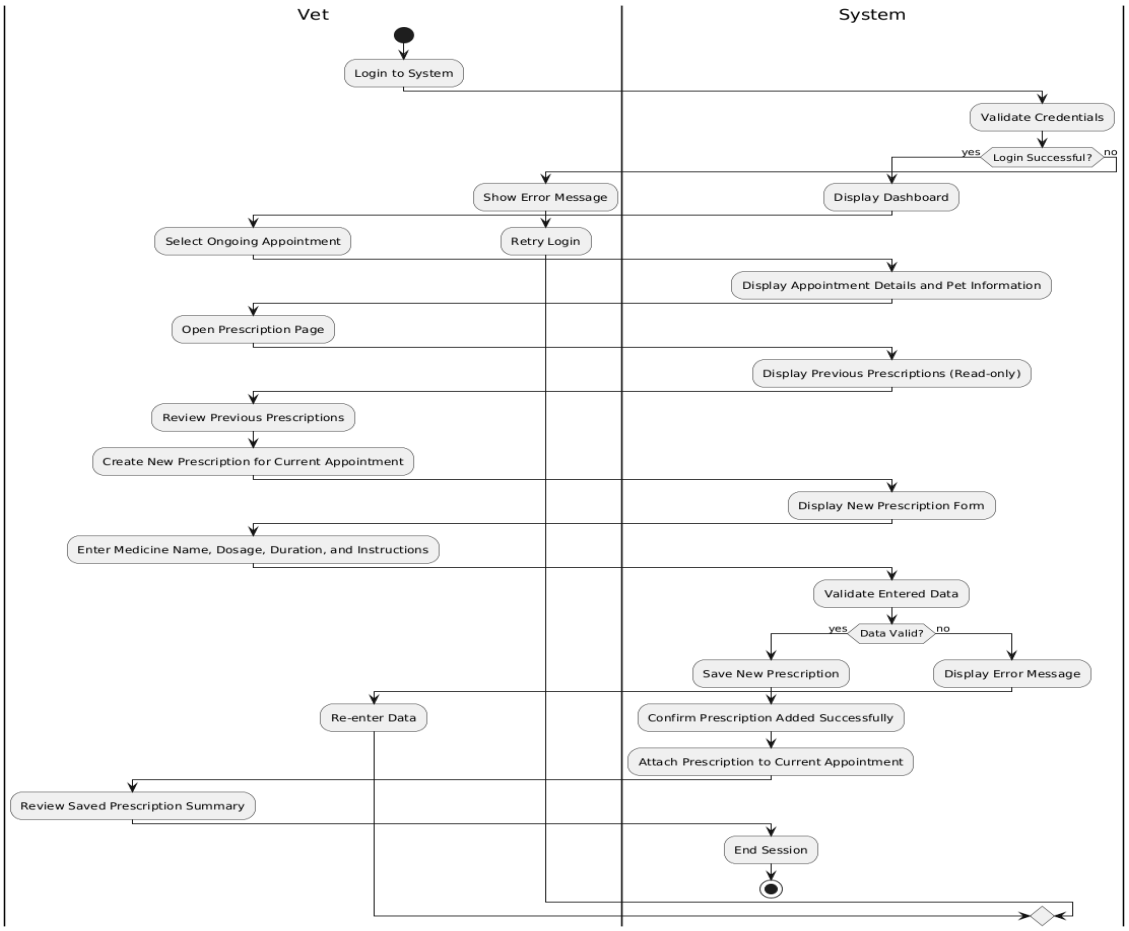

Figure 5.4.8 – Vet Adds Prescription Record

Vet Adds Vaccination Record

Figure 5.4.9 – Vet Adds Vaccination Record

Pet Owner Registration and Login

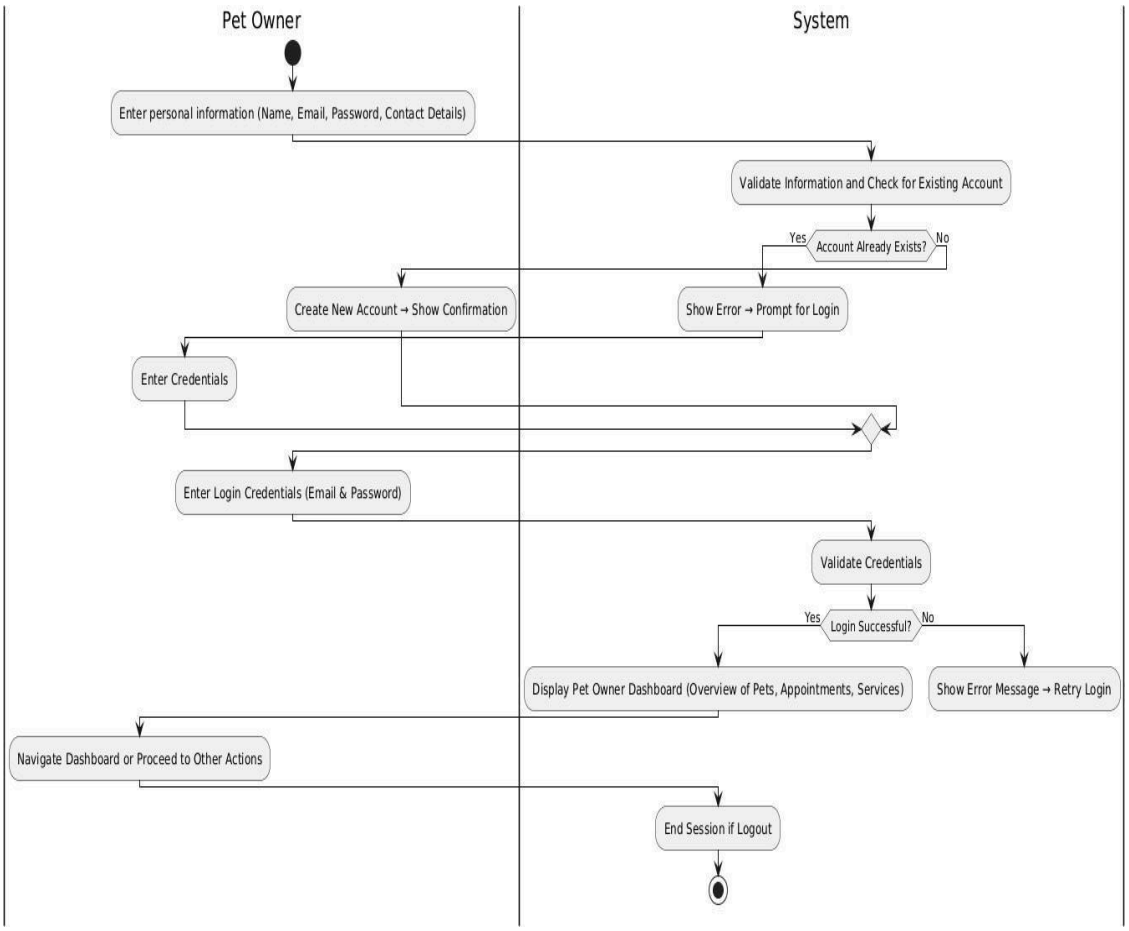

Figure 5.4.10 – Pet Owner Registration and Login

Pet Owner Shop Pet Items

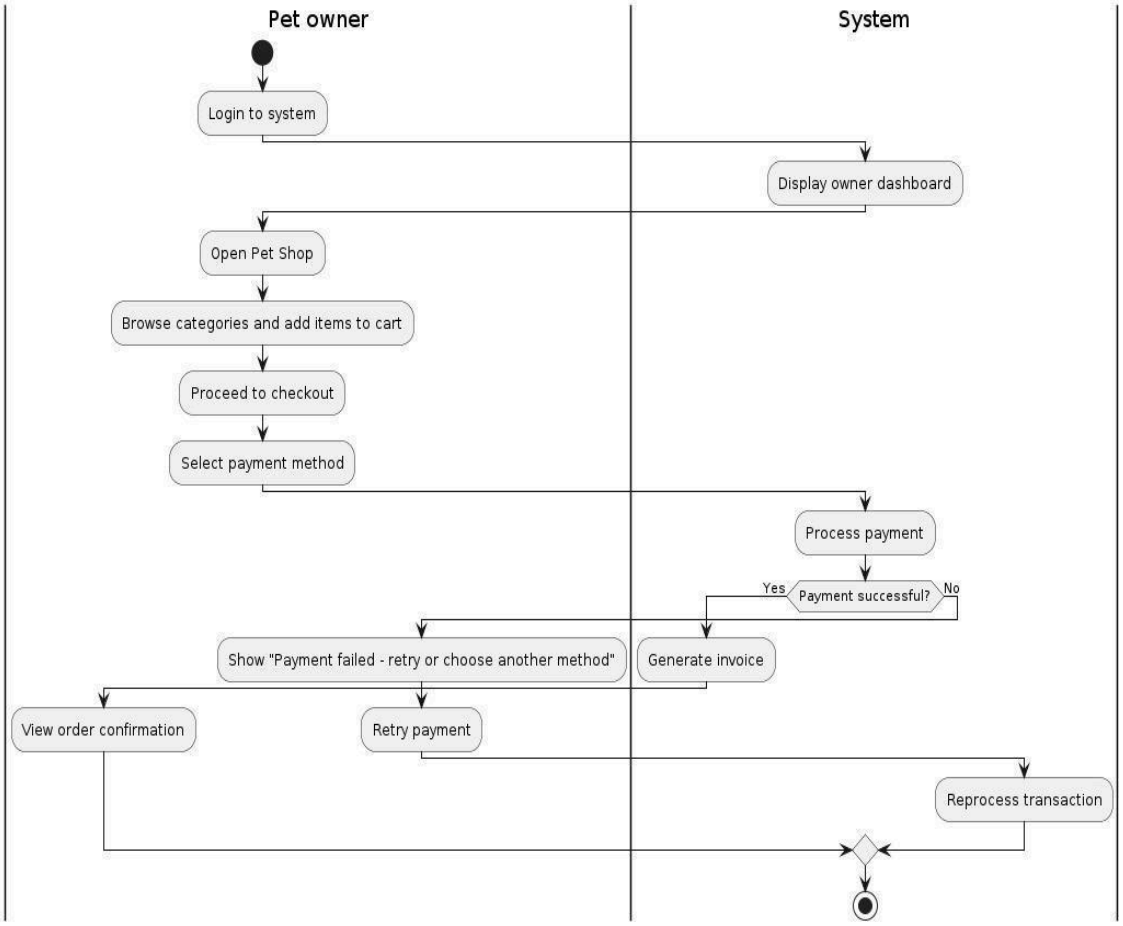

Figure 5.4.11 – Pet Owner Shop Pet Items

Pet Owner Manage Pet Profile

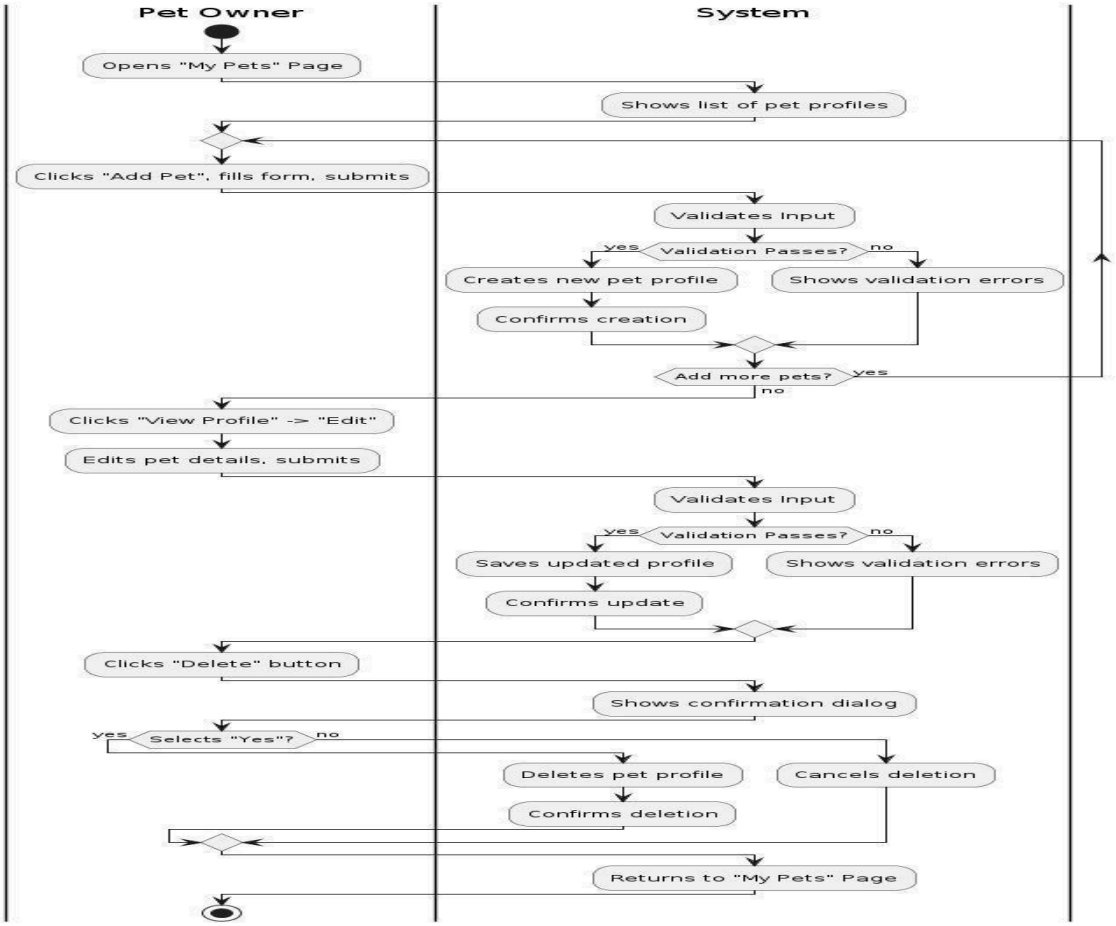

Figure 5.4.12 – Pet Owner Manage Pet Profile

Pet Owner Buy Pets

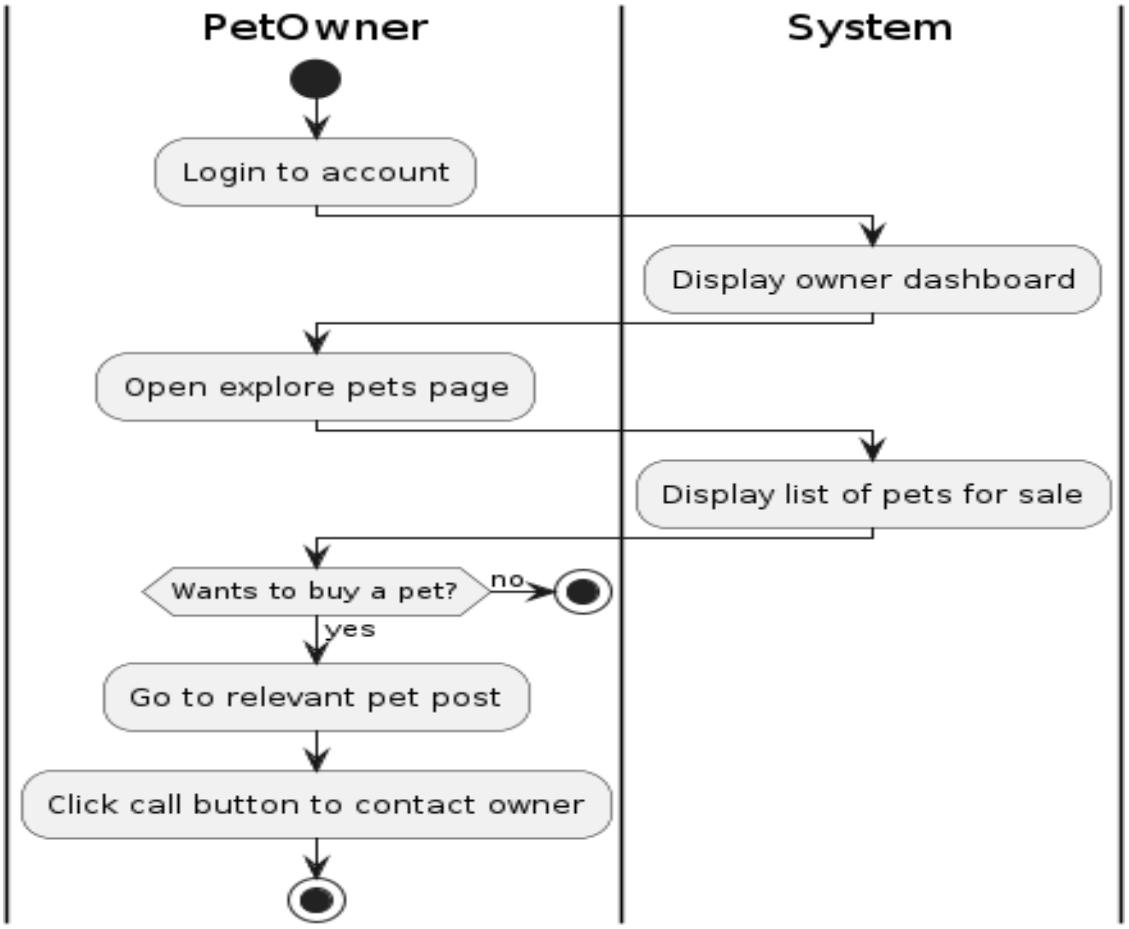

Figure 5.4.13 – Pet Owner Buy Pets

Pet Owner Sell Pets

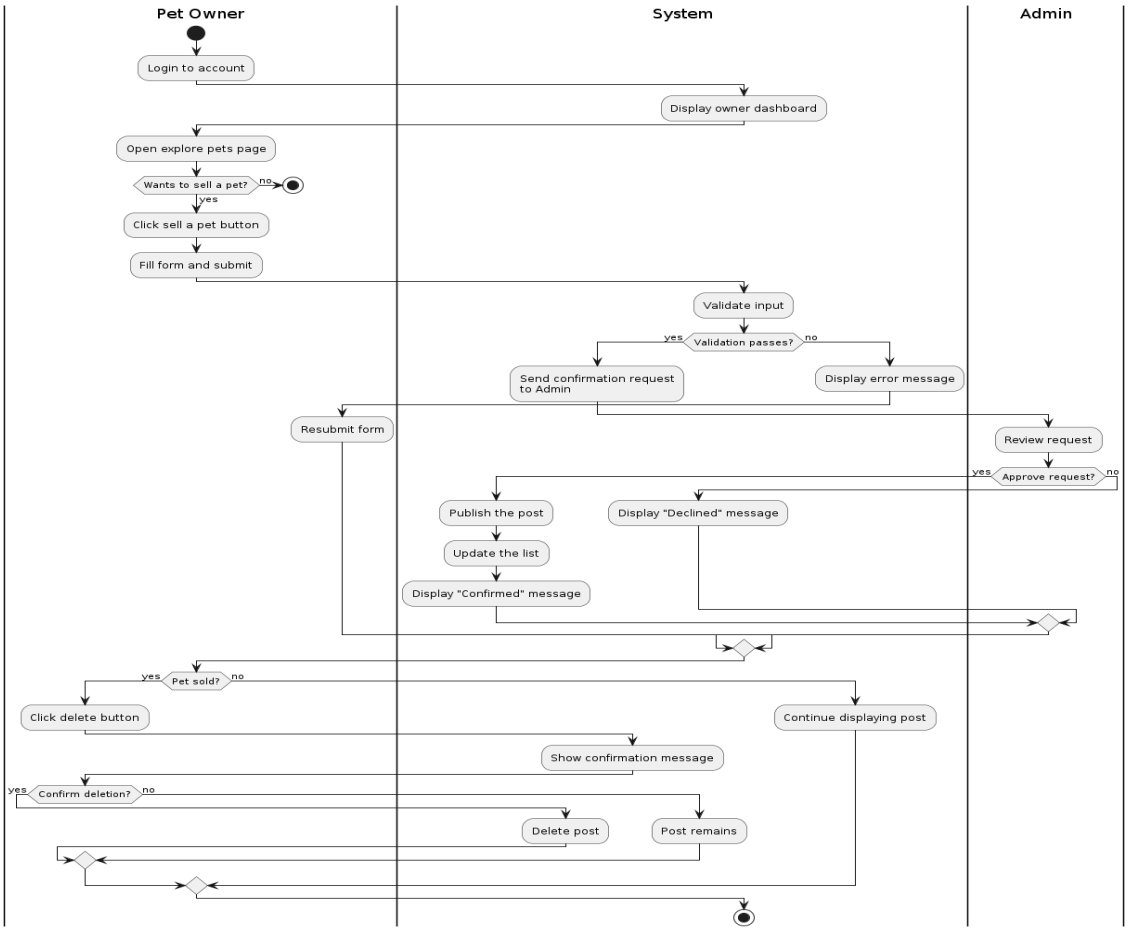

Figure 5.4.14 – Pet Owner Sell Pets

Pet Owner Manage Medical Records

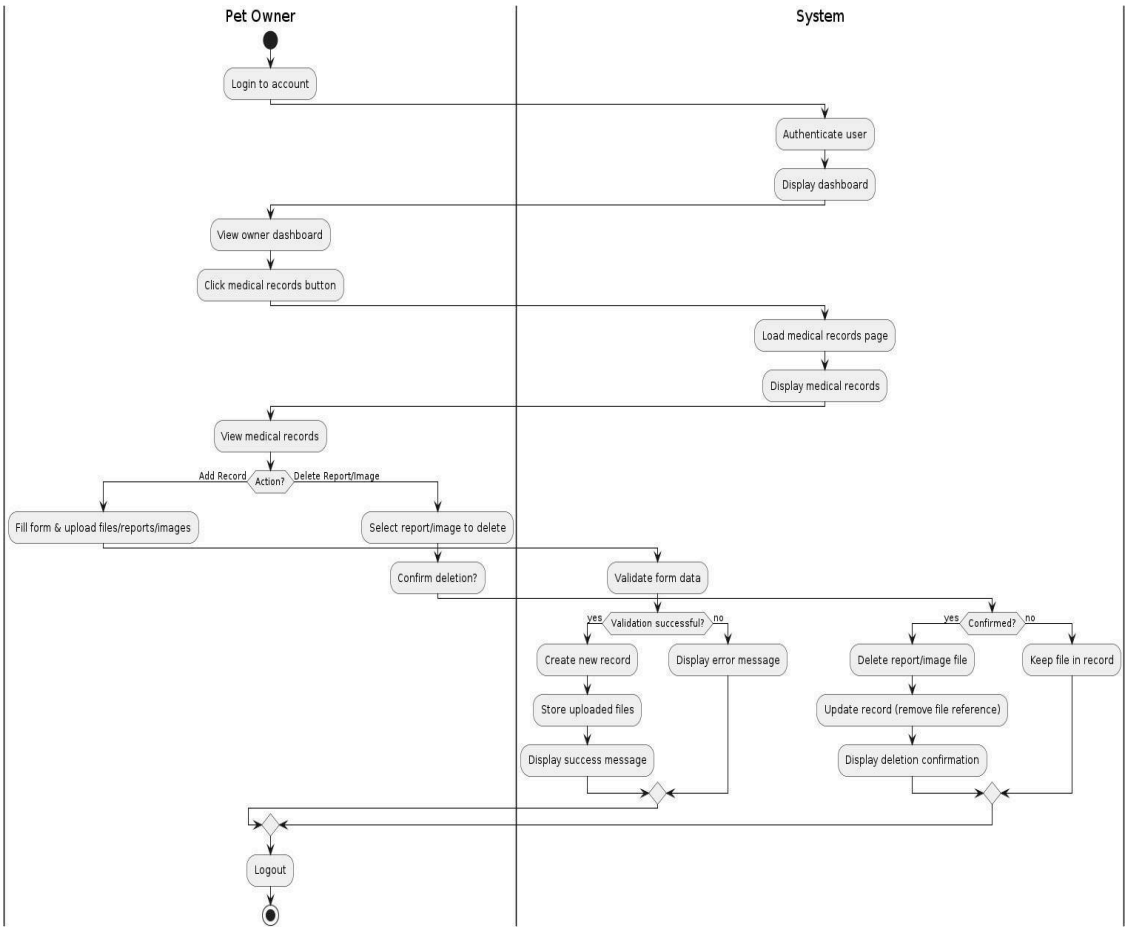

Figure 5.4.15 – Pet Owner Manage Medical Records

Pet Owner Manage Appointments

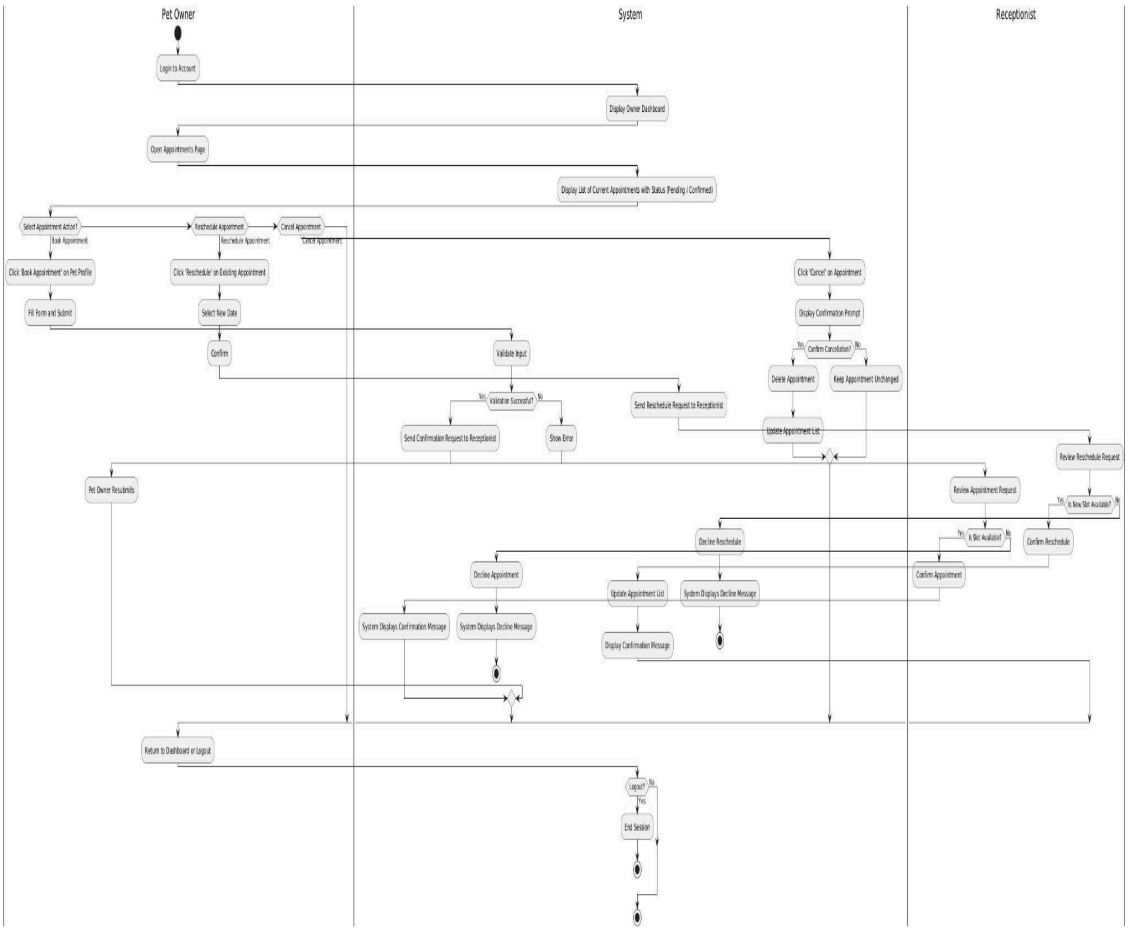

Figure 5.4.16 – Pet Owner Manage Appointments

Pet Owner Reports a Lost Pet

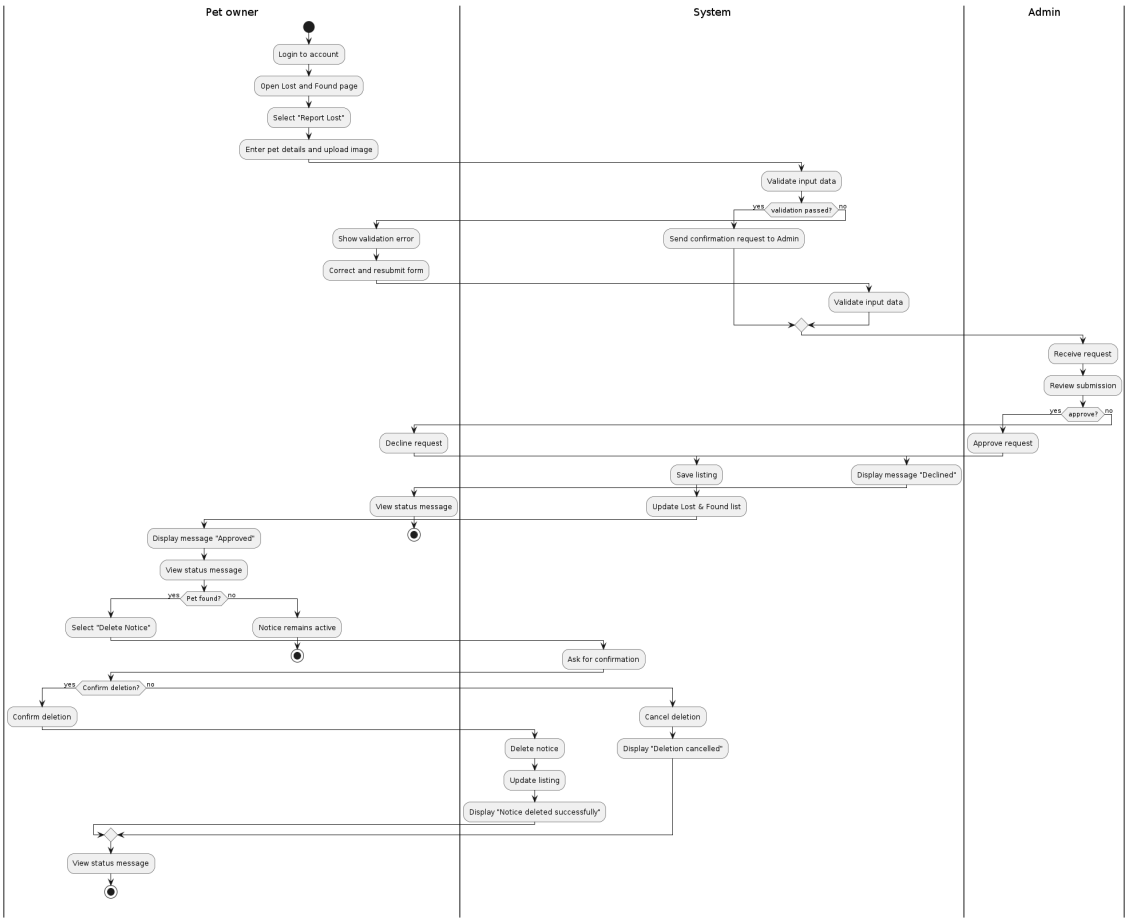

Figure 5.4.17 – Pet Owner Reports a Lost Pet

Pet Owner Reports a Found Pet

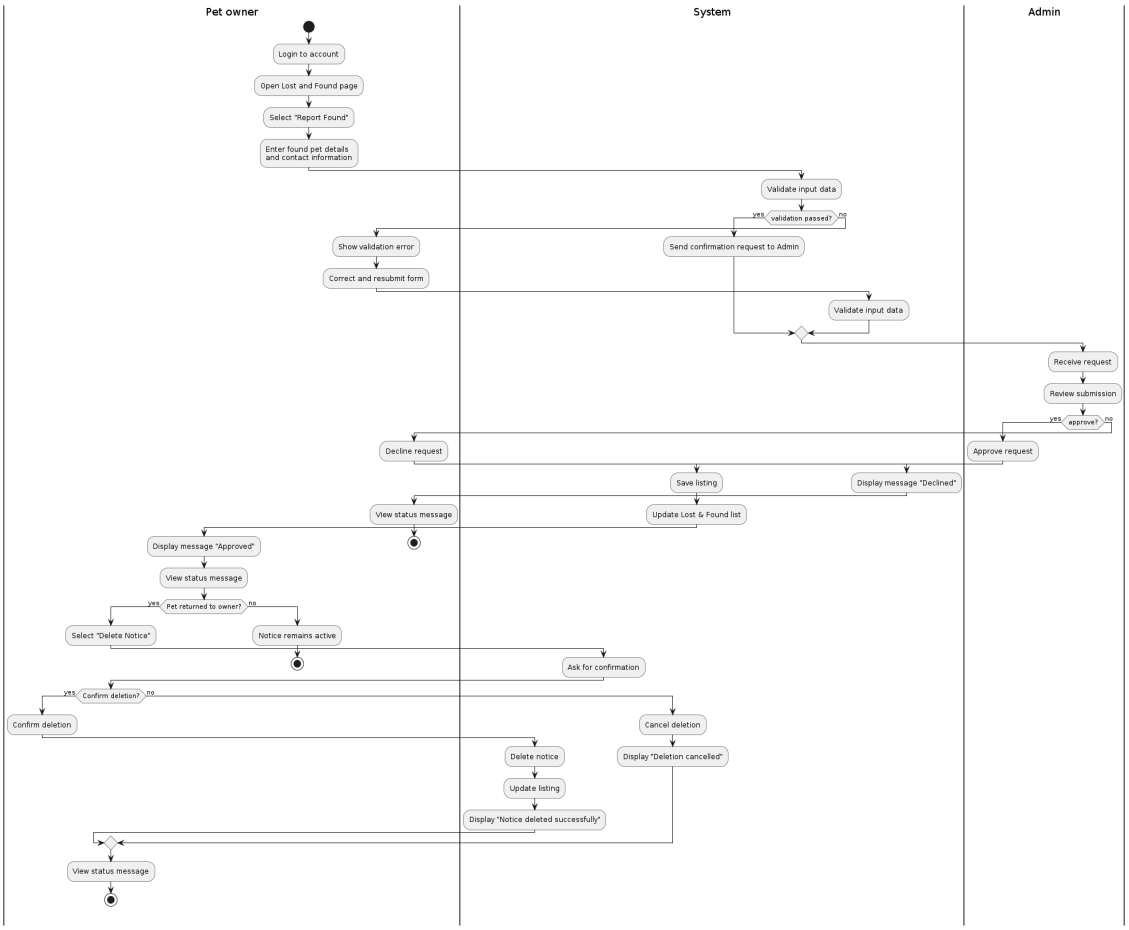

Figure 5.4.18 – Pet Owner Reports a Found Pet

Clinic Manager Manage Appointments

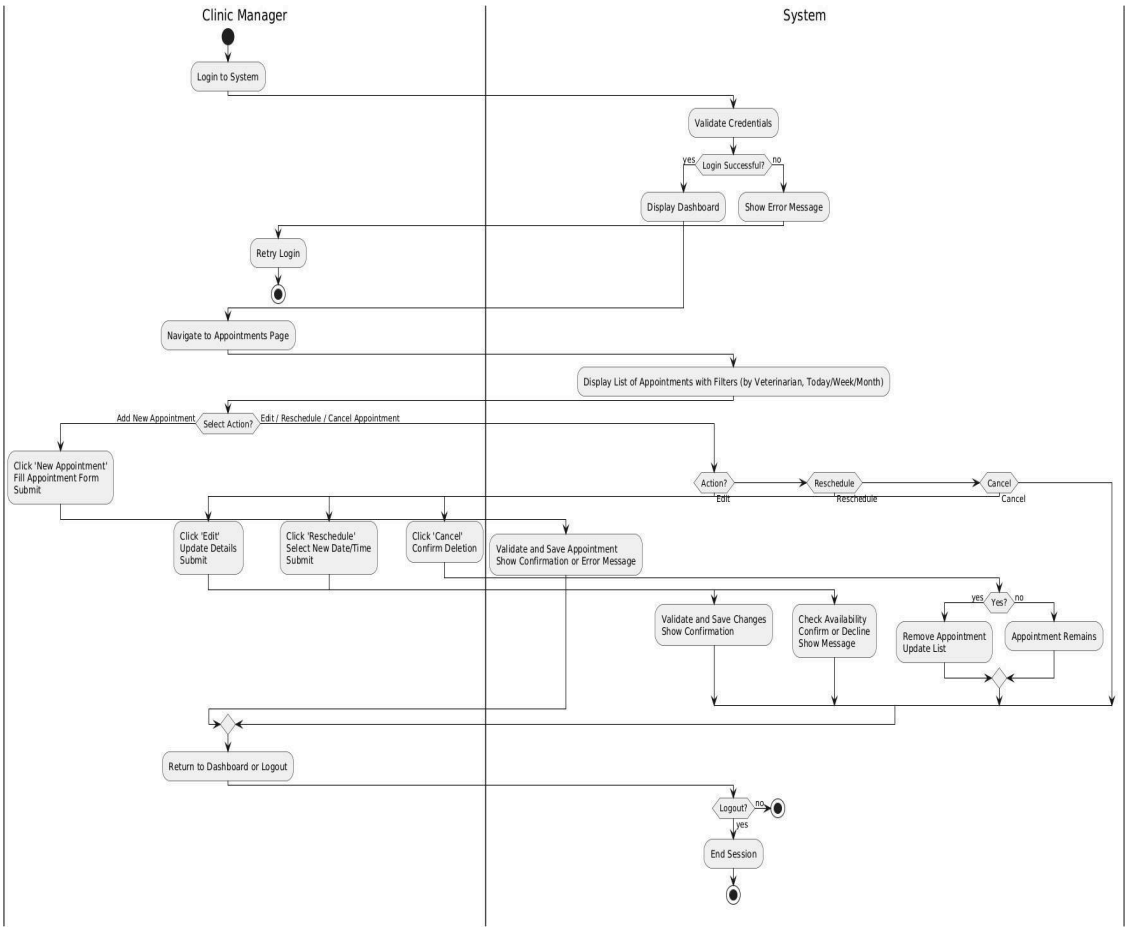

Figure 5.4.19 – Clinic Manager Manage Appointments

Clinic Manager Manage Vets

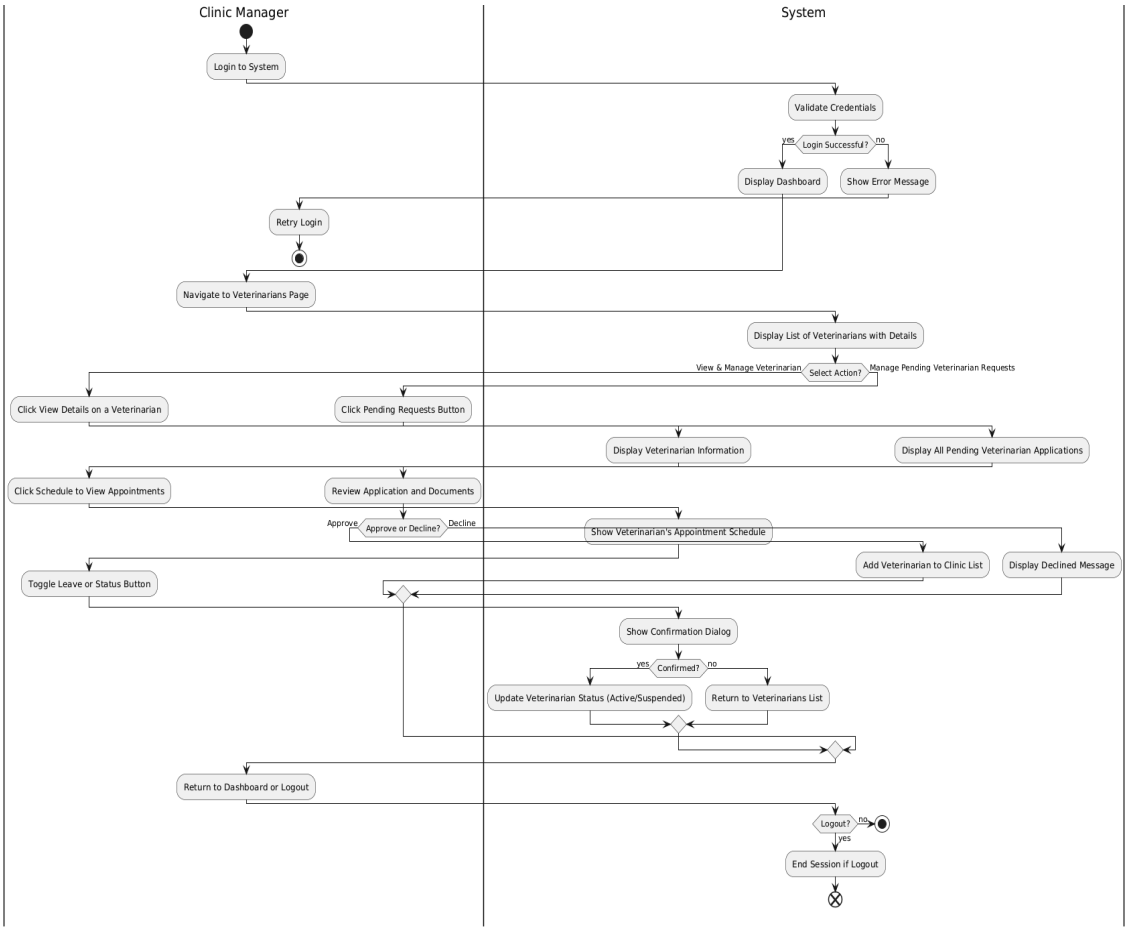

Figure 5.4.20 – Clinic Manager Manage Vets

Clinic Manager Manage Staff

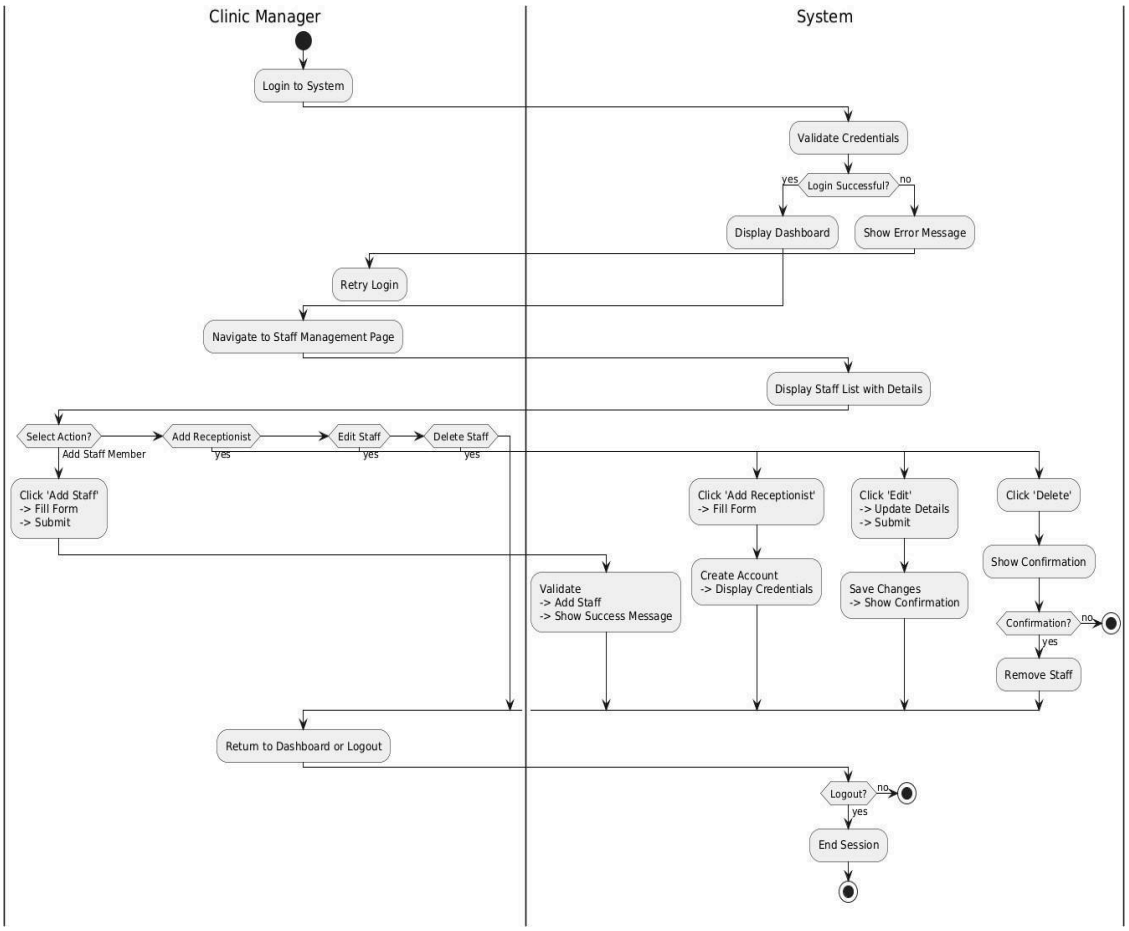

Figure 5.4.21 – Clinic Manager Manage Staff

Clinic Manager Manage Shop Items

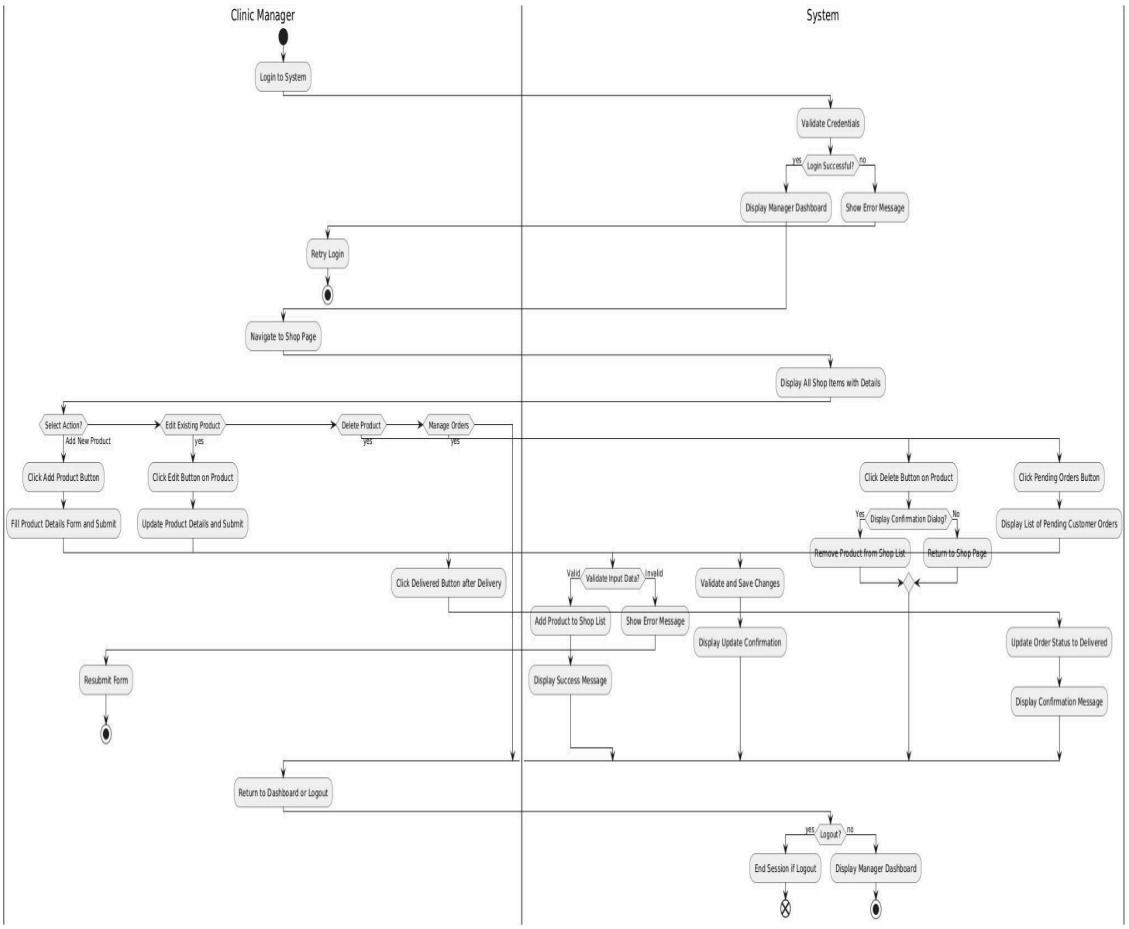

Figure 5.4.22 – Clinic Manager Manage Shop Items

Clinic Manager View Analytics

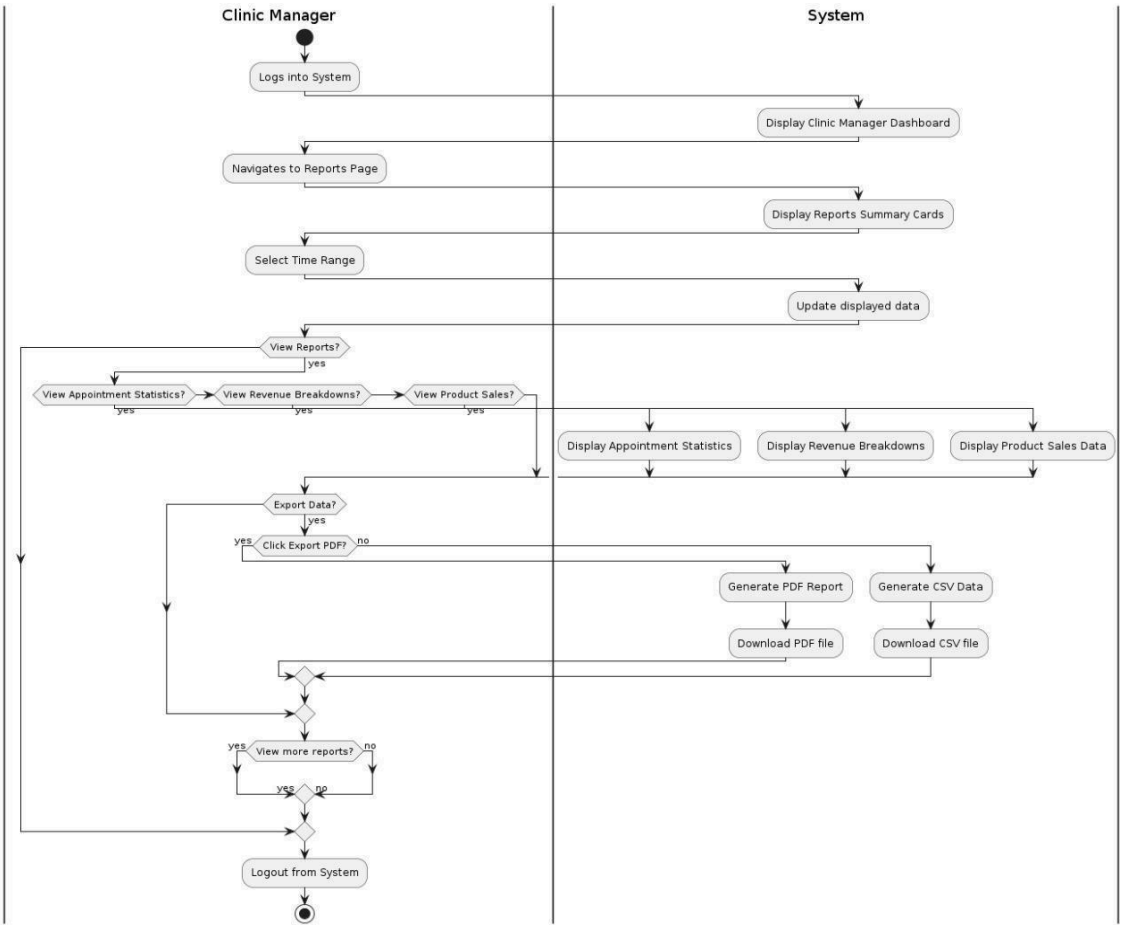

Figure 5.4.23 – Clinic Manager View Analytics

Groomer Service Management

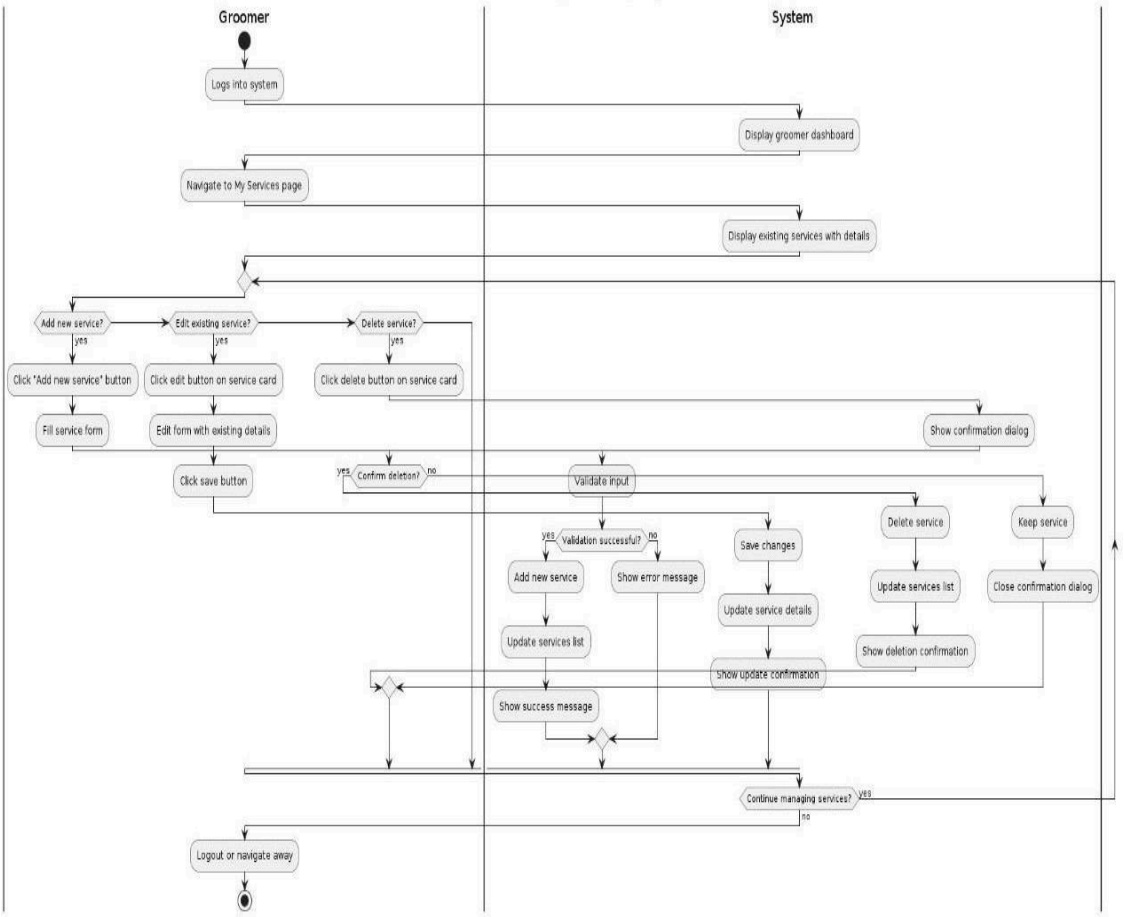

Figure 5.4.24 – Groomer Service Management

Service Provider Availability Management

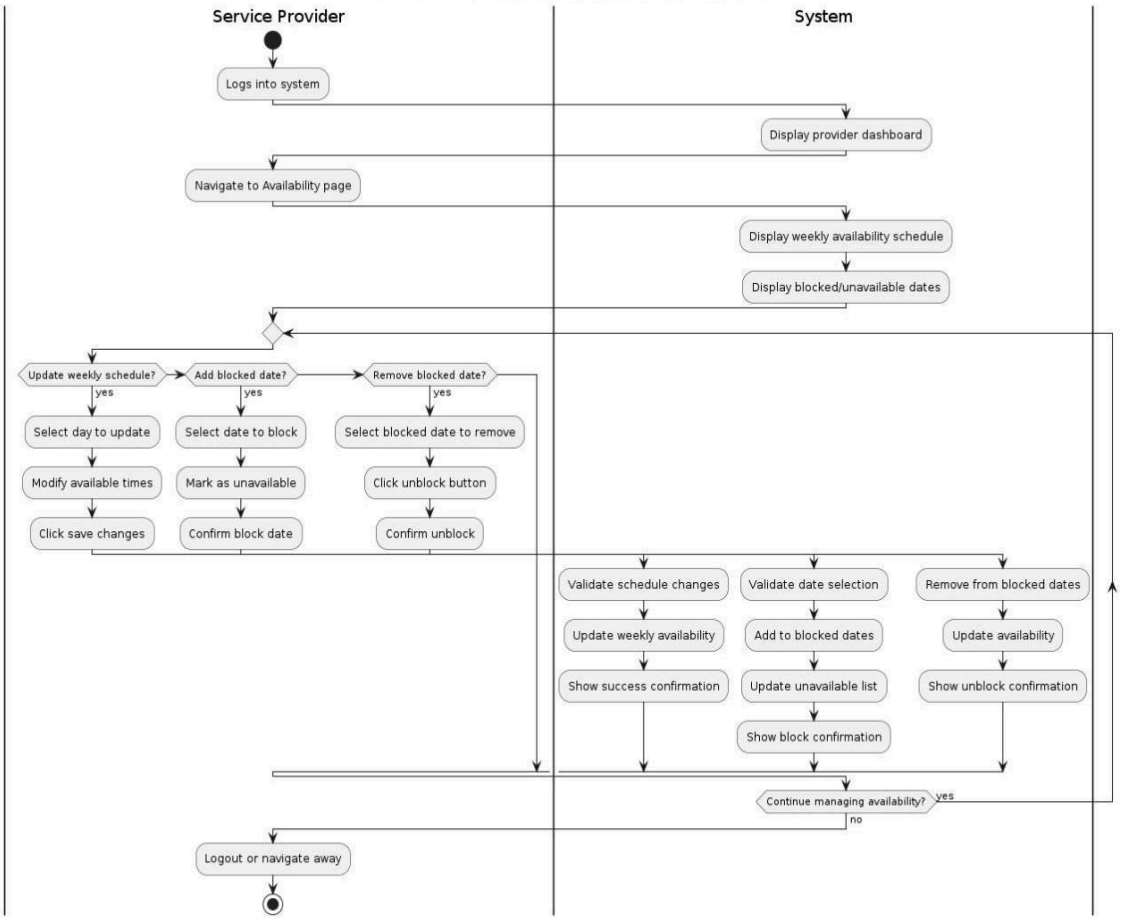

Figure 5.4.25 – Service Provider Availability Management

Register

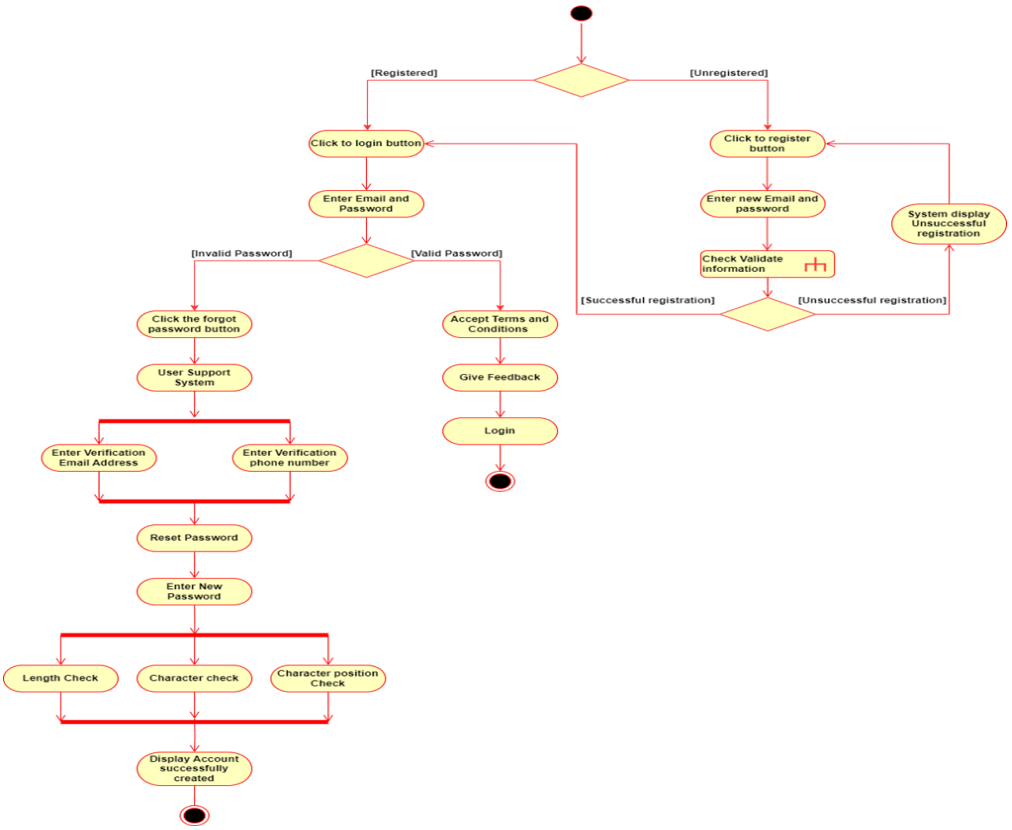

Figure 5.4.26 – Register Activity Diagram

Login As Admin

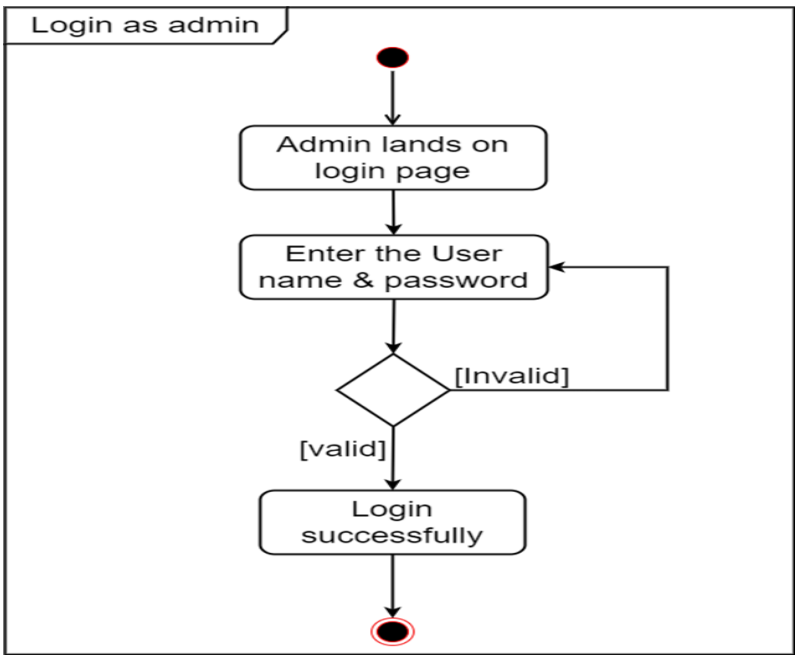

Figure 5.4.27 – Login As Admin Activity Diagram

Login As Manager

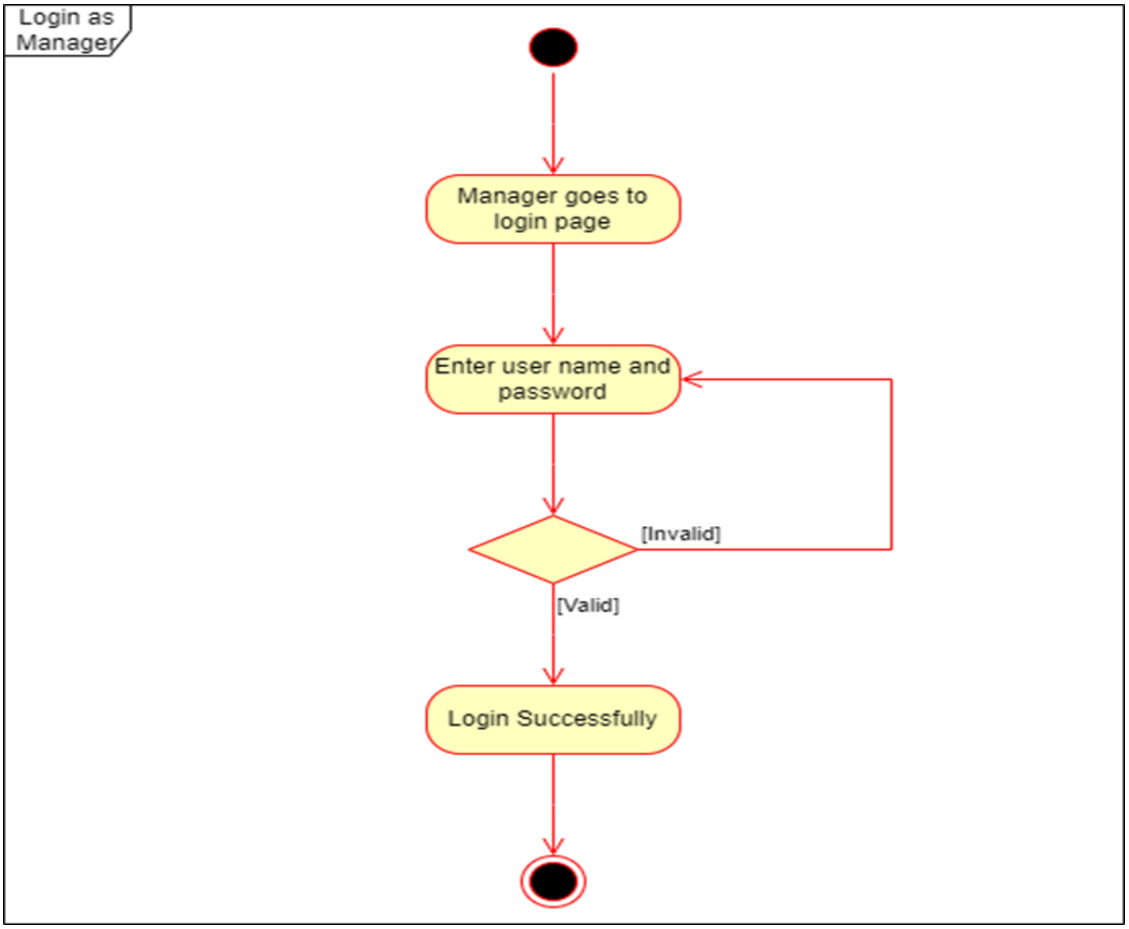

Figure 5.4.28 – Login As Manager Activity Diagram

6. Completeness of the Project

6.1 Functionalities Completed

6.2 Functionalities Yet to Complete

6.3 Individual Contribution of Team Members

Individual Contribution - Member 1
Student ID - 23020725
Student Name - L.M.S Perera
Group ID – 16
Group Name – PETVET System

Name of the component(s)

Role – Pet Owner , Breeder & Admin

• Lost & Found Management System
• Explore Pets (Sell/Adopt Pets)
• Breeding Request Management
• Breeding Pet Management
• Location-Based Filtering
• Leaflet Map Integration
• Admin Pet Listings Review Support
• Pet Shop – Cart & Stripe Payment Integration

Description of the component(s)

• **Lost & Found System** – Designed and developed a complete module for reporting lost and found pets, including form handling, image uploads, and storing structured data such as pet details, contact information, and location coordinates in the database.

• **Interactive Report Viewing Interface** – Implemented a card-based UI to display lost and found reports with pet images, descriptions, location, and contact options for better user interaction.

• **Location-Based Filtering** – Developed distance-based filtering and display nearby reports relevant to users, improving system usability.

• **Advanced Filtering & Sorting** – Implemented search, species filters, and sorting options (newest, oldest, nearest) to enhance user experience.

• **Leaflet Map Integration** – Integrated Leaflet maps to allow users to select and visualize locations, including marker placement and reverse geocoding for address retrieval.

• **Explore Pets Module** – Developed functionality for listing pets for sale or adoption with validation, image uploads, and status-based workflow (pending, approved, rejected).

• **Admin Listing Approval Support** – Implemented logic for admin users to review and approve/reject listings, ensuring only verified listings are visible.

• **Breeding Request Management** – Developed the workflow for sending, receiving, and managing breeding requests with multiple status transitions (pending, approved, declined, completed).

• **Breeding Pet Management** – Enabled breeders to add, update, delete, and manage breeding pets, including availability status and detailed pet information.

• **Form Validation** – Implemented both client-side and server-side validation to ensure data accuracy, prevent invalid inputs, and enhance system reliability.

• **Image Upload & Storage Handling** – Developed functionality for uploading multiple images, storing them efficiently, and retrieving them for display across modules.

• **Database Design Contribution** – Contributed to designing and implementing database tables with proper relationships and constraints for lost & found, listings, breeding, and cart systems.

• **Add to Cart Functionality** – Developed logic to add items to the cart with stock validation and enforced quantity limits (minimum 1, maximum 5 per item) to maintain system constraints.

• **Delivery Charge Calculation** – Developed a delivery calculation system based on user-selected city, including configurable delivery charges and free delivery for orders exceeding Rs. 5000.

• **Stripe Payment Integration** – Integrated Stripe payment gateway using publishable and secret keys to securely process online payments.

Lost & Found – Unit Test Cases

Test Case ID	Test Description	Test Steps	Test Data	Expected Results	Actual Results	Pass/Fail
TL01	Check adding a valid lost pet report	1. Go to Lost & Found page 2. Click "Report Pet" 3. Fill all required fields 4. Upload image 5. Click Submit	Name = Max Type = Lost Species = Dog Location = Colombo Date = Today	Report is successfully created and displayed in Lost list	As expected	Pass
TL02	Check adding report with missing required fields	Follow steps of TL01 but leave required fields empty	Name = (Empty) Species = (Empty)	Error message displayed "Please fill required fields"	As expected	Pass
TL03	Check invalid image upload	Follow steps of TL01 and upload invalid file	File = document.pdf	Error message displayed "Only image files allowed"	As expected	Pass
TL04	Check image size validation	Follow steps of TL01 and upload large image	File size = 7MB	Error message "File size exceeds limit"	As expected	Pass

TL05	Check searching lost pets	1. Go to Lost & Found 2. Enter search term 3. View results	Search = Max	Only matching pets are displayed	As expected	Pass
TL06	Check filtering by species	1. Go to Lost & Found 2. Select species filter	Species = Dog	Only dog reports are displayed	As expected	Pass
TL07	Check contacting pet owner	1. Select a report 2. Click "Contact Owner"	Phone = +94771234567	Contact details are displayed correctly	As expected	Pass
TL08	Check deleting a report	1. Go to My Listings 2. Click Delete 3. Confirm	Report ID = 1	Report is removed successfully	As expected	Pass

Explore Pets – Unit Test Cases

Test Case ID	Test Description	Test Steps	Test Data	Expected Results	Actual Results	Pass/Fail
TE01	Check browsing pets for sale	1. Go to Explore Pets page 2. View default tab	Tab = For Sale	Pet listings for sale are displayed correctly	As expected	Pass
TE02	Check browsing pets for adoption	1. Go to Explore Pets 2. Click "For Adoption"	Tab = For Adoption	Adoption listings are displayed correctly	As expected	Pass
TE03	Check tab switching	1. Go to Explore Pets 2. Switch between tabs	Tabs = Sale / Adoption	Correct content shown for each tab	As expected	Pass

TE04	Check viewing pet details	1. Click "View Details" 2. View modal	Pet = Rocky	Full pet details are displayed	As expected	Pass
TE05	Check search by pet name	1. Enter search 2. View results	Search = Rocky	Only matching pets are shown	As expected	Pass
TE06	Check filter by species	1. Select species filter	Species = Dog	Only dog listings are displayed	As expected	Pass
TE07	Check sorting by price	1. Select sort option	Price = Low → High	Listings sorted correctly	As expected	Pass
TE08	Check contacting owner	1. Click "Contact Owner"	Phone = +94771234567	Contact details are displayed	As expected	Pass
TE09	Check adding new pet listing	1. Click "List a Pet" 2. Fill form 3. Submit	Name = Max Price = 25000	Listing is successfully created	As expected	Pass
TE10	Check validation for empty fields	Follow TE09 with missing fields	Name = (Empty)	Error message displayed	As expected	Pass
TE11	Check editing pet listing	1. Go to My Listings 2. Edit listing	Price = 30000	Listing updated successfully	As expected	Pass
TE12	Check deleting pet listing	1. Go to My Listings 2. Delete listing	Listing ID = 1	Listing removed successfully	As expected	Pass

Breeder Management – Unit Test Case

Test Case ID	Test Description	Test Steps	Test Data	Expected Results	Actual Results	Pass/Fail
TB01	Check viewing breeder dashboard statistics	1. Login as breeder 2. Open dashboard	Breeder ID = 5	Correct statistics are displayed	As expected	Pass
TB02	Check viewing upcoming breeding dates	1. Login 2. View dashboard	Dates = Upcoming	Upcoming breeding dates are displayed correctly	As expected	Pass
TB03	Check adding a new breeding pet	1. Go to Breeding Pets 2. Click "Add Pet" 3. Fill form 4. Submit	Name = Luna Breed = Golden Retriever	Pet is successfully added	As expected	Pass
TB04	Check validation for empty fields	Follow TB03 with missing fields	Name = (Empty)	Error message displayed	As expected	Pass

TB05	Check photo upload for pet	Follow TB03 and upload image	File = luna.jpg	Image uploads successfully	As expected	Pass
TB06	Check editing pet details	1. Go to pet list 2. Click Edit	Description updated	Changes saved successfully	As expected	Pass
TB07	Check activating/deactivating pet	1. Toggle pet status	Status = Active/Inactive	Status updates correctly	As expected	Pass
TB08	Check deleting pet	1. Click Delete 2. Confirm	Pet ID = 10	Pet deleted successfully	As expected	Pass
TB09	Check viewing breeding requests	1. Go to Requests section	Requests = Pending	Requests displayed correctly	As expected	Pass
TB10	Check accepting breeding request	1. Click Accept 2. Fill details 3. Confirm	Date = 2026-05-15	Request status updated to Approved	As expected	Pass
TB11	Check declining breeding request	1. Click Decline 2. Enter reason	Reason = Not available	Request status updated to Declined	As expected	Pass
TB12	Check completing breeding request	1. Click Complete 2. Submit notes	Notes = Successful	Request marked as Completed	As expected	Pass

Admin Module – Unit Test Cases

Test Case ID	Test Description	Test Steps	Test Data	Expected Results	Actual Results	Pass/Fail
TA01	Check viewing admin dashboard statistics	1. Login as admin 2. Open dashboard	Users = 12500	Correct statistics are displayed	As expected	Pass
TA02	Check real-time stats update	1. Open dashboard 2. Add new user	Users = 12501	Stats update automatically	As expected	Pass
TA03	Check viewing all users	1. Go to Manage Users	Users = List	All users are displayed correctly	As expected	Pass
TA04	Check filtering users by role	1. Select role filter	Role = Vet	Only selected role users shown	As expected	Pass
TA05	Check searching users	1. Enter search term	Name = John	Matching users displayed	As expected	Pass
TA06	Check viewing pending requests	1. Open pending tab	Requests = 3	Pending requests displayed	As expected	Pass
TA07	Check approving user	1. Click Approve	User = Sarah	User approved successfully	As expected	Pass
TA08	Check rejecting user	1. Click Reject 2. Enter reason	Reason = Invalid	User rejected successfully	As expected	Pass
TA09	Check viewing clinics	1. Go to Manage Clinics	Clinics = List	Clinics displayed correctly	As expected	Pass

TA10	Check approving clinic	1. Click Approve	Clinic = Happy Paws	Clinic approved successfully	As expected	Pass
TA11	Check generating reports	1. Go to Reports 2. Generate	Report = Users	Report generated successfully	As expected	Pass
TA12	Check downloading CSV report	1. Download report	Format = CSV	File downloaded correctly	As expected	Pass
TA13	Check unauthorized access restriction	1. Login as non-admin 2. Access admin page	Role = User	Access denied	As expected	Pass
TA14	Check updating admin settings	1. Change settings 2. Save	Name updated	Settings saved successfully	As expected	Pass
TA15	Check viewing appointments	1. Go to appointments	Appointments = List	Appointments displayed	As expected	Pass
TA16	Check viewing medical records	1. Go to medical records	Records = List	Records displayed correctly	As expected	Pass
TA17	Check viewing financial data	1. Open finance panel	Revenue data	Financial data displayed	As expected	Pass
TA18	Check session timeout	1. Wait for timeout	Time = 30 mins	User logged out automatically	As expected	Pass
TA19	Check user status change	1. Toggle status	Active → Inactive	Status updated correctly	As expected	Pass

TA20	Check audit trail logging	1. Perform action 2. Check logs	Action = Approve	Action recorded in logs	As expected	Pass

Pet Shop & Payment – Unit Test Cases

Test Case ID	Test Description	Test Steps	Test Data	Expected Results	Actual Results	Pass/Fail
TP01	Check browsing products	1. Go to Pet Shop 2. View products	Products = List	Products displayed correctly	As expected	Pass
TP02	Check filtering by category	1. Select category filter	Category = Dog Food	Filtered products displayed	As expected	Pass
TP03	Check searching products	1. Enter search term	Name = Dog Food	Matching products displayed	As expected	Pass
TP04	Check viewing product details	1. Click product	Product = Premium Dog Food	Product details displayed	As expected	Pass
TP05	Check adding item to cart	1. Click “Add to Cart”	Qty = 1	Item added to cart	As expected	Pass
TP06	Check adding multiple items	1. Add different products	Items = 3	All items added correctly	As expected	Pass
TP07	Check updating quantity	1. Change quantity in cart	Qty = 3	Cart updates correctly	As expected	Pass
TP08	Check removing item from cart	1. Click remove	Item removed	Item removed successfully	As expected	Pass

TP09	Check clearing cart	1. Click clear cart	Cart items = 0	Cart cleared successfully	As expected	Pass
TP10	Check empty cart state	1. Open empty cart	Cart = Empty	Empty message displayed	As expected	Pass
TP11	Check checkout process	1. Click checkout	Items = 3	Checkout initiated successfully	As expected	Pass
TP12	Check delivery calculation	1. Enter location	Distance = 2km	Delivery charge calculated	As expected	Pass
TP13	Check delivery distance limit	1. Enter far location	Distance = 100km	Error message displayed	As expected	Pass
TP14	Check Stripe redirect	1. Click pay	Amount = Rs. 4700	Redirect to Stripe	As expected	Pass
TP15	Check successful payment	1. Enter valid card	C			

Individual Contribution - Member 2
Student ID - 23020032
Student Name - A.G.N.K. Abeywardhana
Group ID – 16
Group Name – PETVET System

Name of the component(s)

Roles: Veterinarian & Receptionist & Pet Owner

• Pet Owner Appointment Booking
• Multi-Clinic Availability Checking

• Veterinarian Assignment & Scheduling
• Pet Owner Appointment Management(Appointment Rescheduling & Cancellation)
• Walk-In Appointment Management
• Appointment Approval & Decline Workflow
• Veterinarian Appointments management
• Medical Records management
• Prescriptions Module & Tracking
• Vaccinations Tracking
• My Packages Management (Groomer)

Description of the component(s)

• **Pet Owner Appointment Booking -** Designed and implemented a comprehensive booking modal integrated into the pet owner's dashboard, enabling pet owners to schedule appointments with interactive calendar widgets, available time slot display, veterinarian selection, and appointment type configuration.

• **Multi-Clinic Availability Checking -** Developed real-time availability validation system preventing double-booking by checking vet schedules, time slot availability, and clinic capacity across multiple veterinary clinics with dynamic slot generation.

• **Appointment Rescheduling Functionality -** Developed appointment rescheduling allowing pet owners to modify appointment dates and times with availability checking, confirmation dialogs, real-time validation, and proper database updates to appointment records.

• **Pet owner Appointment Cancellation -** Implemented appointment cancellation feature with confirmation dialogs, status updates to 'cancelled' and automatic removal from active views.

• **Walk-In Appointment Management -** Developed a complete walk-in appointment registration system allowing receptionists to quickly register new patients, select appointment dates/times, assign veterinarians, and prevent double-booking with real-time availability validation.

• **Appointment Approval & Decline Workflow -** Receptionist-controlled system allowing approval or rejection of pet owner booking requests with automatic status transitions and notifications.

• **Veterinarian Dashboard** – Designed and developed a complete vet dashboard serving as a central hub for veterinarians to manage patient appointments, medical records, prescriptions, and vaccination tracking within their assigned clinic with real-time statistics and status indicators.

• **My Packages Management (Groomer) -** Groomer interface for managing service packages with create, read, update, and delete operations, including pricing, duration, and service management.

• **Form Validation & Data Integrity** – Implemented comprehensive client-side and server-side validation for all forms including required field checks, date/time validation, availability verification, and error messaging to ensure data accuracy and system reliability.

• **Pet Ownership Verification** – Developed security mechanisms to verify pet ownership before allowing booking, medical record access, or appointment management, preventing unauthorized access to other users' data.

• **Advanced Filtering & Sorting** – Implemented search, filters, and sorting options to enhance user experience.

• **Database Design & Optimization** – Contributed to designing and implementing properly normalized database tables (appointments, medical_records, prescriptions, vaccinations, payments) with appropriate relationships, constraints, foreign keys, and indexing for efficient data retrieval.

• **API Development & Endpoints** – Developed multiple API endpoints for appointment operations, medical data retrieval, payment processing, and availability checking with proper request/response handling and error codes.

Individual Contribution - Member 3
Student ID - 23020504
Student Name - H.D.D.H. Liyanage
Group ID – 16
Group Name – PETVET System

Name of the component(s)

Roles: Pet Owner, Clinic Manager, Receptionist

• **All users : Authentication Module :** Implemented the Role-Based Login feature for all users.

• **All Users : Profile Updating, Password Changing & Role Switching** : User Profile Updating, Password changing & Switch between assigned roles of a user easily from the role switcher.

• **Pet Owner : Shop Module** : Designed a clinic wise shops for each clinic to view their products for the Pet Owners

• **Clinic Manager : Veterinarian management** : Manage Veterinarian requests for the clinic, see each veterinarian's Appointments, etc

• **Clinic Manager : Walk-in Appointment Handling** : Gave the ability to add walk-in appointment and handling for the clinic manager as well.

• **Clinic Manager : Staff management** : Clinic Manager can manage their non system access staff, add more staff & add receptionists.

• **Clinic Manager : Shop management** : Can add, edit & delete clinic's shop products.

• **Clinic Manager : Clinic report generation** : Ability to generate weekly, monthly & annual reports.

• **Clinic Manager : Clinic profile, Clinic availability, Shop & Delivery settings** : Manager can configure the clinic's name, location, address, phone. Also the clinic's weekly schedule and appointment slot duration is configurable. The manager can configure the delivery rules fees as well.

• **Receptionist : Payments Processing & Recording** : Billing process & Invoice generation ability was implemented for the receptionist after the appointment is completed by the veterinarian.

• **Appointment Calendar & Filtering** : Built a nice looking calendar themed appointment viewer for the ease of appointment handling for the Clinic manager & Receptionist.

• **Notification system implementation** : Implemented the notification system that notifies the pet owners about the appointment confirmations/declines.

Member	Index No.	Functionalities Completed	Contribution (%)
Perera M.S.	23020725	• CRUD for Pet Listings and Pet selling • CRUD for Lost and found • CRUD for Breeder management	25%
H.D.D.H Liyanage	23020504		25%
A.G.N.K. Abeywardhana	23020032		25%
W.H.A. Sithummal	23021012		25%